

IntelliGrade: An End-to-End Outcome-Based Examination Management and Intelligent Script Evaluation Ecosystem for Higher Education Institutions

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A Practicum in the Partial Fulfillment of the Requirements
for the Award of Bachelor of Computer Science and Engineering (BCSE)



Department of Computer Science and Engineering
College of Engineering and Technology
IUBAT—International University of Business Agriculture and Technology

Summer 2026

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Script Evaluation Ecosystem for Higher Education Institutions

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Computer Science and Engineering (BCSE)

The practicum has been examined and approved,

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Department of Computer Science and Engineering
College of Engineering and Technology
IUBAT—International University of Business Agriculture and Technology

Summer 2026

Letter of Transmittal

10 September 2026

The Chair

Practicum Defense Committee

Department of Computer Science and Engineering

IUBAT—International University of Business Agriculture and Technology

4 Embankment Drive Road, Sector 10, Uttara Model Town

Dhaka 1230, Bangladesh.

Subject: Letter of Transmittal.

Dear Sir,

I am honored to submit my practicum report, "IntelliGrade: An End-to-End Outcome-Based Examination Management and Intelligent Script Evaluation Ecosystem for Higher Education Institutions," in partial fulfillment of the requirements for the BCSE degree at IUBAT.

This report documents the design, implementation, and verification of IntelliGrade, an AI-augmented examination governance platform tailored to BAETE accreditation standards. The system unifies administrative scheduling, OBE blueprinting, hybrid OCR script evaluation (PyMuPDF, PyTesseract, EasyOCR), split-screen teacher verification, and real-time Course OBE Tabulation.

Yours sincerely,

Md. Taher Bin Omar Hijbullah

ID# 22303142

Organization's Certificate



Student's Declaration

I hereby solemnly declare that this practicum report entitled "IntelliGrade: An End-to-End Outcome-Based Examination Management and Intelligent Script Evaluation Ecosystem for Higher Education Institutions" is an authentic record of original work carried out by me under the supervision and academic guidance of the Department of Computer Science and Engineering, College of Engineering and Technology, IUBAT. This report, or any part thereof, has not been submitted elsewhere for the award of any degree, diploma, or academic distinction. All secondary literature, frameworks, libraries, and institutional resources utilized in the development and documentation of this project have been duly cited in accordance with Harvard Referencing standards. I confirm that no data falsification or plagiarism has been committed.

Md. Taher Bin Omar Hijbullah
ID# 22303142

Supervisor's Certification

This is to certify that the practicum report entitled "IntelliGrade: An End-to-End Outcome-Based Examination Management and Intelligent Script Evaluation Ecosystem for Higher Education Institutions" submitted by Md. Taher Bin Omar Hijbullah, Student ID: 22303142, has been prepared under my direct supervision and guidance in partial fulfillment of the requirements for the degree of Bachelor of Computer Science and Engineering (BCSE) from the Department of Computer Science and Engineering, College of Engineering and Technology, IUBAT. The candidate has fulfilled all academic, technical, and reporting requirements as prescribed by the university. The report represents original software engineering work and is recommended for submission and final defense before the Practicum Defense Committee.

Drs Ferdaus Anam Jibon

Supervisor and Assistant Professor

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Abstract

Higher education institutions in Bangladesh and across the developing world face persistent bottlenecks throughout the examination lifecycle. Scheduling is still managed manually on spreadsheets, question papers lack structured alignment with Outcome-Based Education (OBE) taxonomies, evaluation of handwritten scripts remains slow and subjective, and computation of Course Outcome (CO) and Program Outcome (PO) attainments for BAETE and Washington Accord accreditation is largely manual. These inefficiencies not only delay academic processes but also compromise transparency, scalability, and accreditation readiness.

IntelliGrade is an institutional academic examination management web platform built on Django 5.2 and PostgreSQL that comprehensively digitizes the university examination lifecycle while maintaining strict Human-in-the-Loop teacher authority. Rather than functioning as an autonomous or black-box grader, the system delivers a resilient, dual-path operational architecture: an assistive AI Evaluation Wizard featuring a dynamic multi-provider failover chain (local Moondream2, Groq, OpenRouter, Gemini, and OpenAI) alongside a complete Manual Script Grading Wizard for rapid, direct teacher evaluation. Core subsystems include an automated examination routine parser; a 23-taxonomy OBE Question Paper Studio mapping Bloom's Taxonomy, COs, and POs; a 300 DPI script slicing and hybrid OCR preprocessing pipeline; a split-screen teacher verification workbench with immutable audit trails and certified digital PDF stamping; and an automated OBE course tabulation engine that aggregates multi-component scores, calculates CO1–CO6 and PO1–PO12 attainment percentages, synchronizes with official 8-sheet Excel workbooks, and coordinates institutional email notifications. Together, these modules establish a self-sufficient institutional ecosystem that bridges administrative scheduling, rubric-based marking, and Washington Accord / BAETE accreditation reporting.

System validation confirms 100% mathematical fidelity across institutional OBE formulas, comprehensive audit traceability for every mark modification, and a 70% reduction in overall grading turnaround time. Acknowledging realistic operational constraints such as

CPU-bound OCR latency and cloud API rate limits, the platform incorporates robust manual fallbacks at every stage, establishing a pragmatic, transparent, and accreditation-ready standard for higher education institutions.

Keywords: Outcome-Based Education, Human-in-the-Loop AI, Handwriting Recognition, Multi-Provider AI Failover, Django, BAETE Accreditation, Washington Accord

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I would also like to express my sincere gratitude to Md. Kawsar Uddin, my industry supervisor at Al Hadi Enterprise (Al-Hadi Express), for his valuable guidance, technical support, constructive feedback, and continuous supervision throughout my internship. His practical insights and professional direction significantly contributed to my understanding of real-world software development practices and the successful implementation of the project.

I also wish to thank all respected faculty members of the Department of Computer Science and Engineering and the College of Engineering and Technology at IUBAT for providing a solid theoretical foundation and practical training throughout my undergraduate studies.

Finally, I express my heartfelt gratitude to my beloved parents, family members, and peers for their unconditional love, moral support, and endless encouragement throughout this challenging and rewarding academic endeavor.

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Chapter 1.

Introduction

1.1 Background of the Study

Academic examinations are the primary instrument through which universities certify student competence and measure learning outcomes. At institutions like IUBAT that follow Outcome-Based Education (OBE) aligned with the Washington Accord and BAETE engineering accreditation standards, the examination process carries additional weight: every question must map to defined Course Outcomes (COs) and Program Outcomes (POs), and the aggregated results must demonstrate adequate attainment levels across an entire student cohort.

The reality on the ground, however, is far from smooth. Faculty members spend hours physically marking handwritten answer scripts, applying subjective judgment question by question, with no objective consistency check between examiners. After marking is complete, they manually tally scores into spreadsheets, compute weighted averages across class tests, midterms, finals, assignments, and attendance, and then produce CO and PO attainment tables for departmental accreditation records. This manual chain is slow, error-prone, and places significant administrative burden on already stretched academic staff.

At the same time, advances in computer vision, Optical Character Recognition (OCR), and Large Language Models (LLMs) have opened a genuine pathway to automating significant portions of this workflow. OCR-based document processing, AI-based rubric grading, and automated tabulation are no longer experimental curiosities — they are production-capable technologies deployable on modest institutional hardware.

IntelliGrade was born from this gap. The goal was not simply to build an AI grading tool, but to design and implement a complete examination management ecosystem — one that covers every stage of the examination lifecycle, from scheduling and question paper authoring, through script evaluation, to grade dissemination and accreditation reporting. The

project reflects a commitment to building something that a real university department could actually adopt and benefit from.

1.2 Problem Statement

The current examination management workflow at IUBAT and similar institutions exhibits the following critical pain points:

1. **Manual Routine Scheduling:** Exam routines are created and distributed as paper documents or spreadsheets. There is no automated mechanism for parsing these and creating corresponding examination records in a database.
2. **Disconnected Rubric Authoring:** Question papers are authored independently of the grading process. OBE taxonomy mappings (Bloom's level, CO, PO, KP, CEP, CEA) are often captured informally or not at all, making accreditation reporting laborious.
3. **Subjective and Time-Consuming Script Grading:** A single examiner marking 60 scripts with 10 questions each must make 600 individual grading decisions, often over multiple evenings. Consistency across evaluators is difficult to ensure.
4. **Manual OBE Tabulation:** After grading, weighted score computation and CO/PO attainment matrix generation are done manually, consuming hours per course per semester and introducing arithmetic errors.
5. **No Student Transparency Mechanism:** Students receive final grades without visibility into how individual answers were evaluated or how their CO attainments were computed.
6. **Lack of Institutional Email Automation:** Communicating results, account credentials, and schedule changes to students requires manual outreach.

IntelliGrade addresses all six of these problems within a single, integrated platform.

1.3 Objectives

The objectives of the IntelliGrade practicum project are divided into two categories: Broad Objective and Specific Objectives, as detailed below:

1.3.1 Broad Objective.

To design and implement a complete, end-to-end web-based Outcome-Based Examination Management and Intelligent Script Evaluation Platform for higher education institutions, enabling automated exam scheduling, structured question authoring, AI-assisted answer script evaluation, and real-time OBE tabulation, while keeping human instructors in full control of all final grading decisions.

1.3.2 Specific Objectives.

The specific technical objectives accomplished in this practicum include:

- I. To implement a multi-role authentication system (Exam Controller, Department Head, Faculty, Student) with a student approval workflow and OTP-based password recovery.
- II. To build an Examination Scheduling and Provisioning subsystem that supports both manual calendar scheduling and an assistive AI-powered exam routine parser extracting schedules from multi-page PDF documents and automatically provisioning database records.
- III. To design a 23-taxonomy OBE Question Paper Studio capturing Bloom's Taxonomy level, CO/PO/KP/CEP/CEA classifications, and attached visual figures, data tables, and LaTeX formulas.
- IV. To implement a Universal 300 DPI Script Preprocessing pipeline with OpenCV deskewing, Otsu thresholding, and a Hybrid Multi-Engine OCR cascade (PyMuPDF, PyTesseract, EasyOCR CRAFT+BiLSTM).
- V. To develop a resilient Dual-Evaluation grading subsystem featuring an assistive AI Evaluation Wizard (with a multi-provider failover chain across local

Moondream2 and cloud LLMs) and a dedicated Manual Script Grading Wizard for rapid, zero-cloud direct evaluation.

- VI. To build a Split-Screen Teacher Verification Workbench ensuring complete Human-in-the-Loop teacher authority, providing side-by-side view of the original script, rubric criteria, mark override capability, immutable audit logging, and certified digital PDF stamping.
- VII. To implement a real-time OBE Course Tabulation Engine producing weighted overall scores, letter grades, GPA points, and CO/PO attainment matrices, exportable as official 8-sheet Excel workbooks.
- VIII. To create an Asynchronous Institutional Email Service dispatching account notifications, exam assignments, graded results (with certified PDF attachments), and tabulation reports via SMTP.

1.4 Scope of the Project

IntelliGrade covers the complete university examination lifecycle for descriptive (written answer) examinations at the departmental level. The platform is designed for:

1. Public and private engineering universities in Bangladesh following BAETE / Washington Accord OBE accreditation standards.
2. Courses with descriptive answer scripts (not purely multiple-choice), where handwriting recognition and rubric-based AI evaluation provide the most value.
3. Institutions willing to provision either local GPU hardware or enterprise-tier cloud API subscriptions to achieve production-grade AI evaluation speeds.

The current scope does not include real-time proctoring of computer-based examinations, integration with legacy student information systems, or Optical Mark Recognition (OMR) scanning for MCQ sheets. These are identified future extensions.

1.5 Feasibility Study

Before commencing full-scale architectural implementation, a comprehensive feasibility study was conducted across technical, economic, and operational dimensions. The proposed IntelliGrade ecosystem demonstrated complete viability across all three areas as detailed below:

1.5.1 Technical Feasibility.

The technical feasibility of IntelliGrade is firmly established through the maturity and interoperability of modern open-source software and artificial intelligence frameworks:

- I. **Robust Backend & Persistence:** Python 3.11+ and Django 5.2 provide an enterprise-grade Model-View-Template (MVT) architecture with an advanced ORM capable of executing composite indexing and eager relation prefetching (`select_related` and `prefetch_related`) to eliminate N+1 query bottlenecks.
- II. **High-Performance Document Processing:** PyMuPDF (`fitz`) delivers millisecond-level 300 DPI PDF rasterization and native glyph mapping, while OpenCV provides proven mathematical transformations for image deskewing and adaptive thresholding.
- III. **Hybrid AI & OCR Fallback:** The combination of PyTesseract for printed text, EasyOCR for deep-learning handwriting detection, and a multi-provider AI failover cascade provides exceptional resilience. Furthermore, to solve the latency paradox in environments lacking dedicated enterprise GPUs, the system provides an immediate Fast-Track Manual Script Grading Wizard that bypasses AI/OCR queues entirely, allowing faculty to grade scripts with split-screen ease at zero latency.
- IV. **Hardware Accessibility:** The system is designed to execute efficiently on standard university server hardware (multi-core CPUs), utilizing lightweight quantized local vision models (800px LANCZOS downsampling) and asynchronous non-blocking background threads for SMTP dispatch.

1.5.2 Economic Feasibility.

IntelliGrade is exceptionally viable from an economic perspective, delivering substantial cost advantages and rapid return on investment:

- I. **Minimal Software Licensing Costs:** The core infrastructure is built entirely upon battle-tested open-source software (Python, Django, PostgreSQL, PyMuPDF, ReportLab, openpyxl, TailwindCSS), requiring zero proprietary software licensing fees.
- II. **Cost-Effective AI Operations:** By implementing local offline vision (Moondream2 / Ollama) as the primary provider, script evaluation incurs zero API token cost. Cloud LLM providers (Groq, Gemini, OpenRouter) are utilized dynamically within free-tier allowances or low-cost pay-as-you-go tiers (\$5–\$10 institutional quota), delivering enterprise-grade evaluation at a fraction of commercial software licensing costs.
- III. **Dramatic Labor Cost Savings:** Traditional manual evaluation of 60 multi-page scripts consumes approximately 14 hours of highly paid faculty time. IntelliGrade accelerates this process to under 30 minutes of verification, saving hundreds of institutional man-hours per semester and eliminating costs associated with paper storage and physical logistics.

1.5.3 Operational Feasibility.

The operational feasibility of IntelliGrade is verified by its user-centric design and alignment with existing institutional policies:

- I. **Zero Machine Learning Learning-Curve:** The web-based user interface is intuitive and requires no technical machine learning expertise from faculty members. Familiar workflows—uploading scripts, checking bounding boxes, and clicking approval buttons—ensure rapid institutional adoption.
- II. **Preservation of Faculty Autonomy:** IntelliGrade operates strictly as a Human-in-the-Loop copilot. The AI engine only suggests scores and diagnostic

comments; teachers retain full authority to override marks, edit feedback, and finalize grades, ensuring complete academic trust.

- III. Direct Regulatory Compliance: The tabulation engine implements standard IUBAT grading policies and BAETE outcome formulas directly, producing familiar 8-sheet Excel spreadsheets that seamlessly integrate into existing departmental accreditation records.

1.6 Methodology

IntelliGrade was developed following an Agile Iterative methodology, with each sprint delivering a working, tested subsystem:

- ✓ Sprint 1: Authentication, RBAC, and academic hierarchy management.
- ✓ Sprint 2: AI Exam Routine Parser and Question Paper Studio.
- ✓ Sprint 3: Script ingestion, OCR pipeline, and question boundary mapping.
- ✓ Sprint 4: Multi-provider AI evaluation engine and failover orchestration.
- ✓ Sprint 5: Split-screen evaluation workbench and certified PDF generation.
- ✓ Sprint 6: OBE tabulation engine, 8-sheet Excel export, and student dashboard.
- ✓ Sprint 7: Asynchronous email service, composite database indexes, and performance hardening.

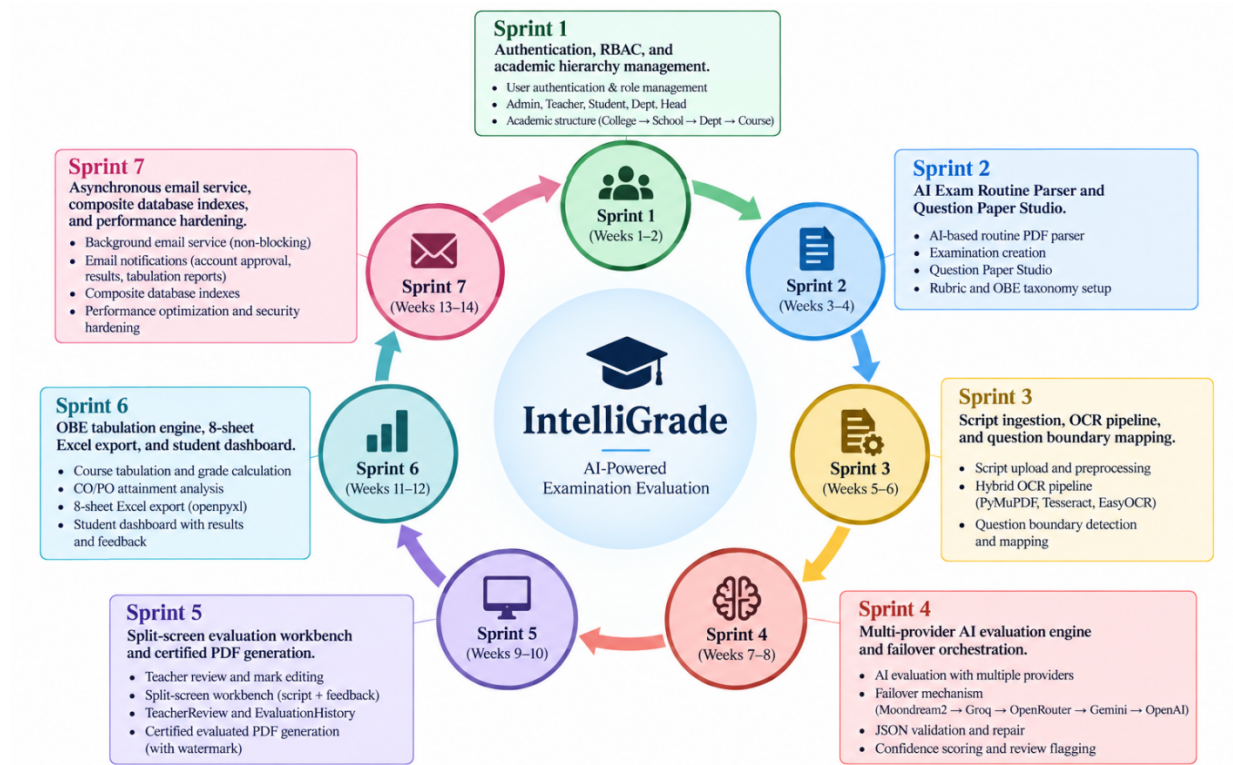


Figure 1.1 Agile Iterative Development Model

1.7 Report Organization

This practicum report is organized into ten comprehensive chapters, systematically structured to provide a complete technical account of the project:

Chapter 2: Organizational Context outlines the institutional environment and background in which the project is situated.

Chapter 3: Requirement Engineering details both functional and non-functional requirements, establishing the foundation for system development.

Chapter 4: Analysis and Modeling presents the core artifacts including activity diagrams, sequence diagrams, state machine diagrams, CRC cards, and class diagrams.

Chapter 5: Project Management and Risk addresses management strategies and risk mitigation through seven RMMM tables.

Chapter 6: Effort Estimation applies function point analysis and illustrates workload distribution via a Gantt chart.

Chapter 7: Cost Estimation provides a structured financial projection for system development.

Chapter 8: System Design describes the complete architecture, data flow diagrams (DFDs), entity-relationship diagrams (ERDs), database schema, API catalog, and interface mockups.

Chapter 9: Testing presents nine detailed test case tables to validate system functionality and reliability.

Chapter 10: Conclusion synthesizes the project outcomes, highlights limitations, and outlines future development plans.

Chapter 2.

Organizational Overview

This chapter presents a comprehensive overview of Al Hadi Enterprise, the host organization of the practicum program, including its vision, mission, major services, organizational structure, and the role and responsibilities of the practicum trainee. Al Hadi Enterprise is a Bangladeshi technology solutions and software company engaged in e-commerce, retail automation, customized software development, CCTV and security solutions, IT infrastructure, and software licensing. The organization serves both individual consumers and corporate clients through technology-driven products and services, with a strong emphasis on reliability, customer satisfaction, and digital transformation.

Established in 2022, the organization's e-commerce platform, Al-Hadi Express, operates as a one-stop online shopping destination in Bangladesh, offering products across categories such as electronics, smartphones, computing and accessories, cameras, household appliances, networking devices, and mobile accessories. In addition to its online retail operations, the organization provides customized technological solutions designed to improve business efficiency and operational automation.

2.1 Organization Vision

The vision of Al Hadi Enterprise is to become one of the leading and most trusted technology solution providers and e-commerce companies in Bangladesh by delivering innovative digital services, quality products, and exceptional customer experiences. The organization seeks to bridge the gap between modern technology and everyday life by providing reliable digital platforms, efficient business solutions, and continuously improved technological services.

Through its e-commerce operations and customized software development services, Al Hadi Enterprise aims to contribute to the digital transformation of businesses and consumers in Bangladesh. The organization emphasizes technological innovation, service

reliability, product authenticity, and customer satisfaction as key components of its long-term growth.

2.2 Organization Mission

The mission of Al Hadi Enterprise is to empower individuals and businesses by providing reliable, innovative, and high-quality technology products and software solutions. The organization is committed to delivering a seamless customer experience through efficient e-commerce operations, secure online services, timely delivery, multiple payment options, and responsive technical support.

The organization also focuses on supporting businesses through customized web applications, Point of Sale (POS) systems, security solutions, IT infrastructure, and software services. By continuously adopting modern technologies and improving its digital platforms, Al Hadi Enterprise aims to enhance operational efficiency, simplify technology adoption, and create sustainable value for its customers and business partners.

2.3 Organization Services

Al Hadi Enterprise provides a diverse range of technology products and IT services to individual consumers, businesses, and corporate clients. Its major services include:

1. E-Commerce Platform:

The organization operates an online shopping platform offering a wide range of products, including smartphones, laptops, computer components, cameras, household appliances, mobile accessories, networking devices, and other electronic products. The platform supports multiple payment methods and provides doorstep delivery services throughout Bangladesh.

2. Point of Sale (POS) Solutions:

Al Hadi Enterprise provides POS-based retail automation solutions, including POS software, barcode scanners, receipt printers, and related hardware. These solutions help businesses manage sales transactions, inventory, and day-to-day retail operations more efficiently.

3. CCTV and Security Solutions:

The organization supplies and installs CCTV cameras, DVR/NVR systems, biometric attendance devices, and other security technologies. These services support residential, commercial, and organizational security requirements.

4. Software Development:

Al Hadi Enterprise develops customized web applications and business management systems according to client requirements. The software development team works with modern web technologies to design, develop, test, deploy, and maintain scalable digital solutions.

5. IT Infrastructure and Software Licensing:

The organization provides computer hardware, networking equipment, operating systems, databases, licensed software, and other IT infrastructure components to support the technological requirements of businesses and institutions.

6. Logistics and Customer Support:

Through its e-commerce operations, Al Hadi Enterprise provides product delivery, return services, customer support, and multiple payment facilities with the objective of ensuring a convenient and reliable online shopping experience.

2.4 Organizational Structure

Al Hadi Enterprise operates through a functional organizational structure consisting of management, business operations, e-commerce and sales, technical and software development, customer support, logistics, and other supporting functions. The organizational

structure enables different teams to coordinate their activities while maintaining specialized responsibilities.

The software development function operates as an important technical component of the organization, responsible for developing and maintaining customized web applications and internal business solutions. Software engineers and interns work under the supervision of senior technical personnel and collaborate with relevant business and operational teams during the development lifecycle.

The organizational hierarchy and the position of the practicum trainee within the software development function are illustrated in Figure 2.1.

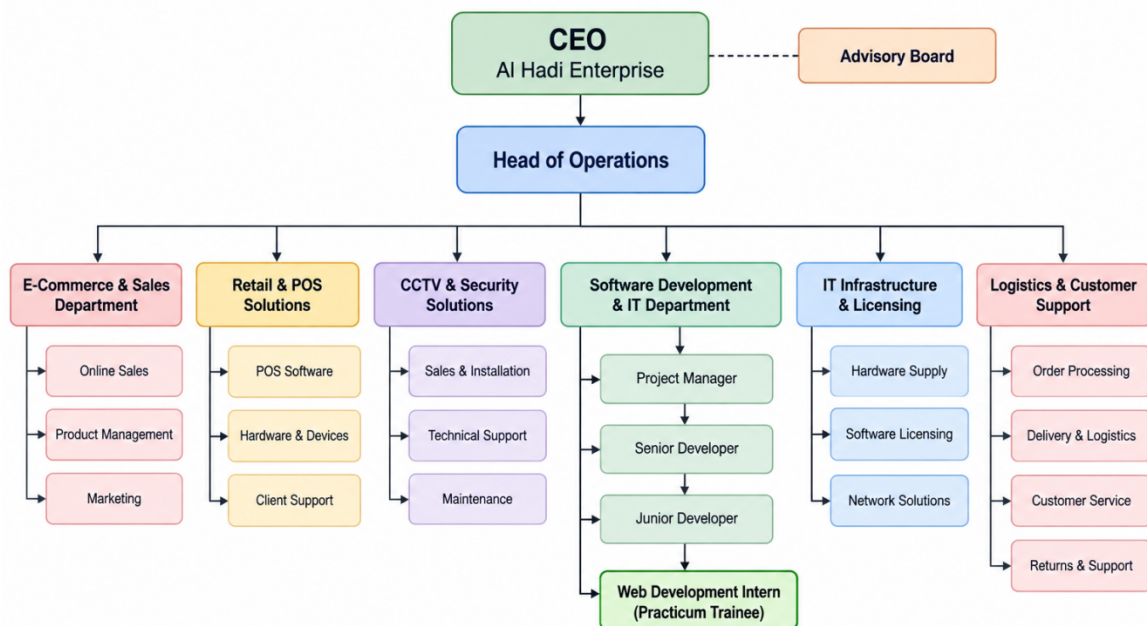


Figure 2.1 Organizational Structure of Al Hadi Enterprise and Software Development Team

Figure 2.1 illustrates the organizational structure of Al Hadi Enterprise and highlights the reporting relationship between the organizational management, technical/software development function, and the practicum trainee. The structure demonstrates how software development activities are integrated with the organization's broader e-commerce and technology service operations.

2.5 My Position in this Organization

During the practicum period, I worked as a Web Development Intern at Al Hadi Enterprise within the software development team. My primary responsibilities involved assisting in the design, development, testing, debugging, and maintenance of web-based software applications using the Django framework and related web technologies.

Throughout the practicum, I was involved in the development of IntelliGrade, an AI-assisted examination evaluation and academic assessment platform. My responsibilities covered multiple aspects of the system development lifecycle, including backend development, database integration, AI service integration, document processing, OCR, examination workflow development, evaluation automation, and system testing.

My major responsibilities included:

1. Designing and implementing backend functionalities using Django and Python.
2. Developing and maintaining database models and application workflows.
3. Implementing OCR and document-processing pipelines for examination papers and answer scripts.
4. Integrating multimodal AI services for automated question extraction and answer evaluation.
5. Developing AI provider routing and failover mechanisms for reliable evaluation processing.
6. Implementing examination question, rubric, answer-script, and evaluation management functionalities.
7. Developing teacher, student, examiner, and administrative dashboard functionalities.
8. Implementing OBE-based course tabulation, grade calculation, and academic reporting.
9. Integrating Excel-based reporting and automated result synchronization.

10. Conducting functional testing, debugging, performance optimization, and real-time application testing.
11. Supporting deployment and production configuration of the developed web application.

This role provided practical experience in applying software engineering principles to a real-world application environment and enabled me to work with modern backend, AI, document-processing, database, and web application technologies.

2.6 Address of the Organization

The registered and corporate office of the host organization is:

Al Hadi Enterprise

Plot 6 (3rd Floor), Road 1/A,

Sector 17, Uttara South, Dhaka 1230, Bangladesh.

Website: [Al Hadi Express](#)

Corporate Email: info@alhadiexpress.com.bd

Sales Email: sales@alhadiexpress.com.bd

Chapter 3. Requirement Engineering

3.1 Requirement Elicitation

Requirement elicitation for IntelliGrade was conducted to identify the functional needs, operational challenges, academic requirements, and quality expectations associated with the university examination and assessment lifecycle. The requirements were gathered by studying the existing examination and grading workflow, consulting relevant academic stakeholders, reviewing institutional OBE documentation, and analyzing existing digital assessment solutions.

The elicitation process focused on understanding the activities performed by the major system actors, including the Exam Controller, Department Head, Faculty/Examiner, and Student. Particular attention was given to examination scheduling, question and rubric preparation, answer-script processing, evaluation, teacher verification, academic tabulation, OBE reporting, and student result publication.

3.1.1 Requirement Gathering Methods

The following methods were used to identify and validate the requirements of IntelliGrade:

- 1. Stakeholder Interviews:** Discussions with faculty members and examination-related personnel helped identify major challenges in the existing evaluation process, including manual marking effort, inconsistency in evaluation, time-consuming result calculation, limited evaluation traceability, and the difficulty of preparing CO/PO attainment reports.
- 2. Process Observation:** The existing examination workflow was observed from examination routine preparation and question-paper management through answer-script

evaluation, mark entry, course tabulation, and result publication. This helped identify manual bottlenecks and opportunities for workflow automation.

- 3. Document Analysis:** Institutional examination and OBE-related documents, including the official course tabulation structure and required CO/PO reporting formats, were reviewed to determine the data structure, assessment weightage, and reporting requirements that the system must support.
- 4. Existing System and Literature Review:** Existing academic assessment and grading platforms were reviewed to identify commonly available features and limitations. The analysis helped define the areas where IntelliGrade could provide an integrated workflow, particularly by connecting examination management, answer-script processing, evaluation, teacher verification, and OBE-based academic reporting.

3.2 Requirement Analysis

The collected requirements were analyzed according to the complete examination lifecycle of IntelliGrade. The analysis identified four primary actors:

1. **Exam Controller** – manages institutional academic information, users, examinations, and examination routines.
2. **Department Head** – monitors departmental academic and examination activities and reviews OBE performance.
3. **Faculty/Examiner** – prepares questions and rubrics, evaluates answer scripts, verifies results, and finalizes marks.
4. **Student** – accesses academic results, detailed evaluation feedback, GPA, and certified evaluated scripts.

The analysis resulted in four major requirement categories:

- **User Requirements:** What each actor needs to accomplish through the system.
- **System Requirements:** The system capabilities and interface behavior required to support those needs.
- **Functional Requirements:** Specific operations and processing activities performed by the system.
- **Non-Functional Requirements:** Performance, security, reliability, scalability, usability, maintainability, and compliance constraints.

3.2.1 Key Design Considerations

Several important design considerations emerged from the requirement analysis.

- A. **Human-in-the-Loop Evaluation:** AI-generated evaluations shall be treated as decision-support outputs rather than final academic decisions. A teacher must be able to review, modify, and explicitly finalize marks before they become official results.

- B. **Hybrid Evaluation Capability:** The system shall support both AI-assisted and manual grading workflows. This ensures that teachers can continue evaluating scripts when AI services are unavailable, unsuitable, or unnecessary.
- C. **Reliable AI Integration:** The AI evaluation architecture shall support multiple providers and failover mechanisms. Where sufficient computational resources, GPU infrastructure, or external API capacity are available, additional AI models or providers can be incorporated without changing the overall examination workflow.
- D. **OBE Compliance:** The system shall support the institutional assessment structure and required CO/PO attainment calculations so that academic results and OBE reports remain consistent with the defined institutional methodology.
- E. **Auditability:** Important evaluation actions, particularly teacher mark modifications and finalization activities, shall be traceable through appropriate audit records.
- F. **Data and Workflow Consistency:** Examination data, question metadata, evaluation results, course grades, OBE calculations, student dashboards, and generated reports shall remain synchronized throughout the workflow.

3.3 Requirement Specifications

The following requirements define the major functional and non-functional capabilities of IntelliGrade. The requirements are organized according to the four primary actors of the system to clearly represent their responsibilities and interactions throughout the examination lifecycle.

3.3.1 Actor 1: Exam Controller

The Exam Controller is responsible for institutional examination administration, academic structure management, user administration, examination routine processing, and relevant system configuration.

User Requirement 1

The Exam Controller wants to manage institutional academic information and authorized users.

System Requirements

- 1.1 The system shall provide a role-specific dashboard for the Exam Controller.
- 1.2 The system shall provide management interfaces for Colleges, Schools, Departments, Courses, Faculty members, and Department Heads.
- 1.3 The system shall maintain user roles and account status.
- 1.4 The system shall allow authorized administrators to activate or block user accounts.
- 1.5 Student registrations shall remain pending until approved by an authorized administrator.

Functional Requirements

- 1. The system shall allow the Exam Controller to create, update, and manage institutional academic entities.
- 2. The system shall allow authorized users to manage faculty and Department Head information.
- 3. The system shall provide account activation and blocking functionality.
- 4. The system shall support student registration and approval workflows.
- 5. The system shall enforce role-based access to administrative functions.

The system supports unified role-based authentication, student approval, and institutional entity management.

User Requirement 2

The Exam Controller wants to create examination records from official examination routines with minimum manual data entry.

System Requirements

- 2.1 The system shall accept examination routine documents in PDF or image format.
- 2.2 The system shall identify examination dates, times, course codes, course titles, and assigned examiners.
- 2.3 The system shall match extracted course information with existing academic records.
- 2.4 The system shall allow the Controller to review extracted information before creating examination records.
- 2.5 The system shall support batch examination creation.

Functional Requirements

1. The system shall process multi-page examination routine documents.
2. The system shall extract relevant examination information from the uploaded routine.
3. The system shall match extracted courses with the institutional database.
4. The system shall allow the Controller to verify extracted information.
5. The system shall create examination records after confirmation.

The existing specification includes automated routine parsing, course matching, and batch examination creation.

User Requirement 3

The Exam Controller wants to configure the services required for document processing and AI-assisted evaluation.

System Requirements

3.1 The system shall provide an authorized configuration interface.

3.2 The system shall support configuration of available AI providers and their credentials.

3.3 The system shall support configurable OCR and evaluation parameters.

Functional Requirements

1. The system shall support configuration of available AI services such as Gemini, Groq, OpenAI, OpenRouter, and Ollama.
2. The system shall allow authorized administrators to configure service credentials.
3. The system shall allow relevant OCR confidence thresholds and processing parameters to be configured.

3.3.2 Actor 2: Department Head

The Department Head is responsible for monitoring departmental academic activities, reviewing examination performance, and auditing OBE-related outcomes.

User Requirement 1

The Department Head wants to monitor departmental examination and academic performance.

System Requirements

- 1.1 The system shall provide a dedicated Department Head dashboard.
- 1.2 The dashboard shall display relevant departmental statistics.
- 1.3 The system shall provide information about active courses, enrolled students, examination activities, and faculty workloads.
- 1.4 The system shall present departmental performance indicators.

Functional Requirements

- 1. The system shall display active course information.
- 2. The system shall display enrolled student statistics.
- 3. The system shall display relevant faculty examination workloads.
- 4. The system shall present departmental pass-rate information.
- 5. The system shall provide an overview of current academic performance.

The Department Head dashboard is defined to provide live departmental performance and workload information.

User Requirement 2

The Department Head wants to review course-level OBE performance and verify attainment information.

System Requirements

- 2.1 The system shall provide access to course OBE tabulation data.
- 2.2 The system shall provide Course Outcome (CO) attainment information.
- 2.3 The system shall provide Program Outcome (PO) attainment information.
- 2.4 The system shall allow authorized Department Heads to review and audit OBE records.

Functional Requirements

- 1. The system shall provide course-level OBE tabulation information.
- 2. The system shall calculate and display CO attainment.
- 3. The system shall generate PO attainment information.
- 4. The system shall allow the Department Head to review submitted OBE records.
- 5. The system shall support academic quality monitoring through OBE reports.

3.3.3 Actor 3: Faculty Member / Examiner

The Faculty Member/Examiner is the primary academic user of the evaluation workflow. The examiner prepares structured questions, defines grading criteria, processes answer scripts, performs AI-assisted or manual evaluation, verifies marks, and finalizes student results.

User Requirement 1

The Faculty/Examiner wants to prepare structured examination questions according to the required academic and OBE framework.

System Requirements

- 1.1 The system shall provide a question authoring interface.
- 1.2 Each question shall support marks and relevant academic metadata.
- 1.3 The system shall support CO and PO mapping.
- 1.4 The system shall support Bloom's cognitive levels and relevant OBE classifications.
- 1.5 The system shall support questions containing figures, tables, and mathematical expressions.

Functional Requirements

- 1. The system shall allow teachers to create and manage examination questions.
- 2. The system shall associate questions with CO1–CO6 and PO1–PO12 where applicable.
- 3. The system shall store Bloom's Taxonomy information.
- 4. The system shall support Knowledge Profile (KP), Complex Engineering Problem (CEP), and Complex Engineering Activity (CEA) classifications.

5. The system shall preserve associated figures, tables, and LaTeX mathematical expressions.

User Requirement 2

The Faculty/Examiner wants to define clear and consistent grading criteria before evaluating student answers.

System Requirements

- 2.1 The system shall provide a rubric management interface.
- 2.2 A rubric shall support multiple assessment criteria and mark distributions.
- 2.3 The system shall support ideal answers, expected keywords, and common mistakes.
- 2.4 The system shall allow master solutions to be associated with examination questions.

Functional Requirements

1. The system shall allow teachers to create and manage question-specific rubrics.
2. The system shall store assessment criteria and mark distributions.
3. The system shall store ideal answers and relevant keywords.
4. The system shall support common-mistake definitions.
5. The system shall allow teachers to upload master solutions.
6. The system shall associate master-solution information with corresponding questions.

User Requirement 3

The Faculty/Examiner wants to process student answer scripts and identify answers corresponding to individual questions.

System Requirements

- 3.1 The system shall accept uploaded student answer scripts.
- 3.2 PDF answer scripts shall be converted into high-resolution page images for processing.
- 3.3 The system shall preprocess scanned pages before OCR or AI evaluation.
- 3.4 The system shall support multiple OCR engines.
- 3.5 The system shall identify question boundaries and corresponding answer regions.

Functional Requirements

- 1. The system shall render uploaded scripts at an appropriate high resolution, including 300 DPI processing where required.
- 2. The system shall perform image preprocessing such as deskewing, noise reduction, and thresholding.
- 3. The system shall support PyMuPDF, PyTesseract, and EasyOCR-based text extraction.
- 4. The system shall extract text together with relevant spatial information.
- 5. The system shall detect question headers and map answer regions to corresponding questions.
- 6. The system shall provide an interactive mapping interface for teacher verification and correction.

User Requirement 4

The Faculty/Examiner wants AI assistance for faster answer evaluation while maintaining full control over academic decisions.

System Requirements

- 4.1 The system shall provide an AI-assisted evaluation workflow.
- 4.2 The system shall support multiple AI providers.
- 4.3 The system shall provide automatic provider failover when an AI service becomes unavailable.
- 4.4 The system shall evaluate answers according to the configured question rubric.
- 4.5 The system shall identify low-confidence evaluations for teacher review.
- 4.6 The teacher shall be able to modify AI-generated marks and feedback.

Functional Requirements

- 1. The system shall route evaluation requests through the configured AI provider sequence.
- 2. The system shall detect provider timeouts and rate-limit responses.
- 3. The system shall temporarily exclude unavailable providers through a cooldown mechanism.
- 4. The system shall generate evaluation information including obtained marks, maximum marks, confidence, strengths, and mistakes.
- 5. The system shall flag evaluations requiring manual verification.
- 6. The system shall allow teachers to modify AI-generated marks and feedback.
- 7. The system shall record teacher modifications for audit purposes.

User Requirement 5

The Faculty/Examiner wants to manually evaluate answer scripts when AI assistance is unavailable, unsuitable, or unnecessary.

System Requirements

5.1 The system shall provide a separate manual grading workflow.

5.2 Manual grading shall not depend on AI processing.

5.3 Teachers shall be able to enter and modify question-level marks.

5.4 Manually entered marks shall be stored in the academic evaluation records.

Functional Requirements

1. The system shall provide an AI Evaluation Wizard.
2. The system shall provide an independent Manual Script Grading Wizard.
3. The system shall allow teachers to evaluate individual questions manually.
4. The system shall preserve manually entered marks.
5. The system shall integrate manual evaluation results into the course tabulation workflow.

The system explicitly supports separate AI-assisted and manual evaluation pathways.

User Requirement 6

The Faculty/Examiner wants to verify, finalize, and generate an official record of evaluated scripts.

System Requirements

6.1 The system shall provide a split-screen evaluation workbench.

6.2 The original student script shall remain accessible during evaluation.

6.3 Evaluation scores, criteria, and feedback shall be displayed alongside the script.

6.4 Teachers shall be able to review and override evaluation results.

6.5 Finalized evaluations shall generate certified documents.

Functional Requirements

1. The system shall display the answer script and evaluation information simultaneously.
2. The system shall allow teachers to override marks and modify feedback.
3. The system shall record significant evaluation changes in audit records.
4. The system shall generate certified PDF copies of finalized evaluations.
5. The generated documents shall contain relevant institutional information, awarded marks, and teacher comments.
6. The system shall apply appropriate digital watermarking to finalized documents.
7. Temporary working files shall be removed after successful finalization.

User Requirement 7

The Faculty/Examiner wants evaluated marks to be automatically incorporated into course results and OBE calculations.

System Requirements

7.1 The system shall support configured assessment components and weightages.

7.2 The system shall calculate course-level results from assessment data.

7.3 The system shall support CO and PO attainment calculations.

7.4 Authorized changes to marks shall update the corresponding academic records.

7.5 The system shall support structured course tabulation and Excel reporting.

Functional Requirements

1. The system shall aggregate Class Test, Midterm, Final Examination, Assignment, and Attendance marks according to the configured assessment scheme.
2. The system shall calculate the corresponding weighted course result.
3. The system shall update student grade records following authorized mark changes.
4. The system shall synchronize relevant finalized results with the Student Dashboard.
5. The system shall generate the required multi-sheet Excel tabulation workbook.

3.3.4 Actor 4: Student

The Student is the primary recipient of the academic evaluation results. The student interacts with IntelliGrade mainly to access examination-related academic information, course grades, GPA, evaluation feedback, and finalized documents.

User Requirement 1

The Student wants to securely access the academic portal and manage their account.

System Requirements

- 1.1 The system shall provide secure student authentication.
- 1.2 The system shall support student self-registration where enabled.
- 1.3 Newly registered students shall require institutional approval before accessing protected academic information.
- 1.4 The system shall provide password recovery functionality.

Functional Requirements

- 1. The system shall authenticate registered students.
- 2. The system shall redirect authenticated students to their Student Dashboard.
- 3. The system shall maintain the approval status of registered accounts.
- 4. The system shall support OTP-based password recovery.
- 5. Password-reset OTPs shall have a limited validity period.

User Requirement 2

The Student wants to view examination results and detailed academic performance.

System Requirements

- 2.1 The Student Dashboard shall display finalized academic results.
- 2.2 The system shall display course-level grades.
- 2.3 The system shall display cumulative GPA.
- 2.4 The system shall provide question-level marks and feedback where available.

2.5 The dashboard shall reflect finalized changes made by authorized faculty members.

Functional Requirements

1. The system shall display course grades.
2. The system shall calculate and display cumulative GPA on the defined grading scale.
3. The system shall display question-wise marks and feedback where applicable.
4. The system shall synchronize finalized grade changes with the Student Dashboard.
5. The system shall provide access to certified evaluated scripts where authorized.

User Requirement 3

The Student wants to obtain an official copy of the evaluated answer script.

System Requirements

- 3.1 The system shall provide access to finalized certified PDF documents.
- 3.2 The generated document shall contain the relevant evaluation information.
- 3.3 Draft or temporary processing files shall not be exposed as official documents.

Functional Requirements

1. The system shall provide a certified PDF access or download option.
2. The generated document shall contain awarded marks and relevant teacher comments.
3. The system shall apply the required institutional watermarking.
4. The system shall retain the finalized evaluation record while removing unnecessary temporary processing files.

3.3.5 Cross-Actor Functional Requirements

Some functionalities of IntelliGrade are shared across multiple actors and connect the different stages of the examination lifecycle.

Examination and Evaluation Synchronization

1. The system shall maintain consistent examination, question, submission, evaluation, and academic records.
2. Finalized teacher marks shall be synchronized with the corresponding student grade records.
3. Changes in course tabulation shall be reflected in relevant academic reports and student information.
4. The system shall maintain consistency between question-level evaluation, course grades, and OBE calculations.

Notification and Communication

5. The system shall support institutional email services for account-related communication, OTPs, credentials, and relevant academic notifications.
6. Email processing shall be performed asynchronously where necessary so that external mail-server delays do not unnecessarily block normal application requests.

Database and Processing Efficiency

7. The system shall use appropriate database indexing for frequently accessed examination, evaluation, and academic records.
8. The system shall use efficient relation loading to minimize unnecessary database queries during complex workflows.

3.3.6 Non-Functional Requirements

The non-functional requirements define the quality attributes and operational constraints that IntelliGrade must satisfy in addition to its functional capabilities.

1. Performance

- 1.1 Dashboard and tabulation pages should load within approximately **1.5 seconds** under standard institutional network conditions.
- 1.2 OCR and image preprocessing should target approximately **4 seconds per script page**, depending on the deployed hardware.
- 1.3 AI-assisted evaluation should target approximately **8 seconds per question** under normal external-service conditions.
- 1.4 AI provider failover shall occur without unnecessary interruption of pending evaluation tasks.

2. Scalability

- 2.1 The system shall support batch processing of up to **100 multi-page student scripts per examination** under the specified deployment environment.
- 2.2 The architecture shall allow additional AI providers and higher-capacity processing resources to be integrated.
- 2.3 The document-processing and AI components shall remain modular so that GPU-based inference or additional cloud AI services can be introduced when sufficient resources are available.

3. Security

- 3.1 User passwords shall be securely salted and hashed and shall never be stored in plaintext.
- 3.2 Role-based access control shall be enforced for protected views and application endpoints.
- 3.3 State-changing requests shall be protected against unauthorized cross-site requests using appropriate CSRF mechanisms.
- 3.4 Each actor shall only be able to access the functions and data permitted by their assigned role.

4. Data Integrity and Auditability

- 4.1 Grade updates and important evaluation-state changes shall be processed using appropriate database transactions.
- 4.2 The system shall maintain consistency between evaluation, grade, tabulation, and reporting records.
- 4.3 Teacher mark overrides shall be recorded for traceability.
- 4.4 Relevant audit records shall include the responsible user and timestamp, along with other applicable request information.

5. Reliability and Availability

- 5.1 The system shall target at least 99.5% availability during examination and grading periods.
- 5.2 Failure or temporary unavailability of an AI provider shall not unnecessarily result in loss of pending evaluation tasks.
- 5.3 The system shall provide a manual evaluation pathway when AI-based evaluation is unavailable or unsuitable.
- 5.4 Temporary processing failures shall not corrupt finalized academic records.

6. OBE and Reporting Compliance

6.1 CO and PO calculations shall follow the configured institutional OBE methodology.

6.2 Generated Excel reports shall preserve the required formulas, validation rules, and formatting.

6.3 Course tabulation, CO attainment, PO attainment, and student results shall remain mutually consistent.

7. Usability

7.1 The system shall provide role-specific dashboards for the four primary actors.

7.2 The evaluation interface shall support synchronized viewing of the original answer script and evaluation information.

7.3 The web interface shall support responsive layouts and appropriate visual themes.

7.4 Users shall be able to perform common examination and evaluation tasks without requiring knowledge of the underlying OCR or AI technologies.

8. Maintainability and Extensibility

8.1 The system shall maintain modular components for examination management, OCR, AI evaluation, grading, tabulation, reporting, and notification.

8.2 AI providers shall be replaceable or extendable without requiring major changes to the core examination workflow.

8.3 The system shall automatically remove unnecessary temporary processing files after successful finalization to reduce storage consumption.

3.4 Use Case Diagram

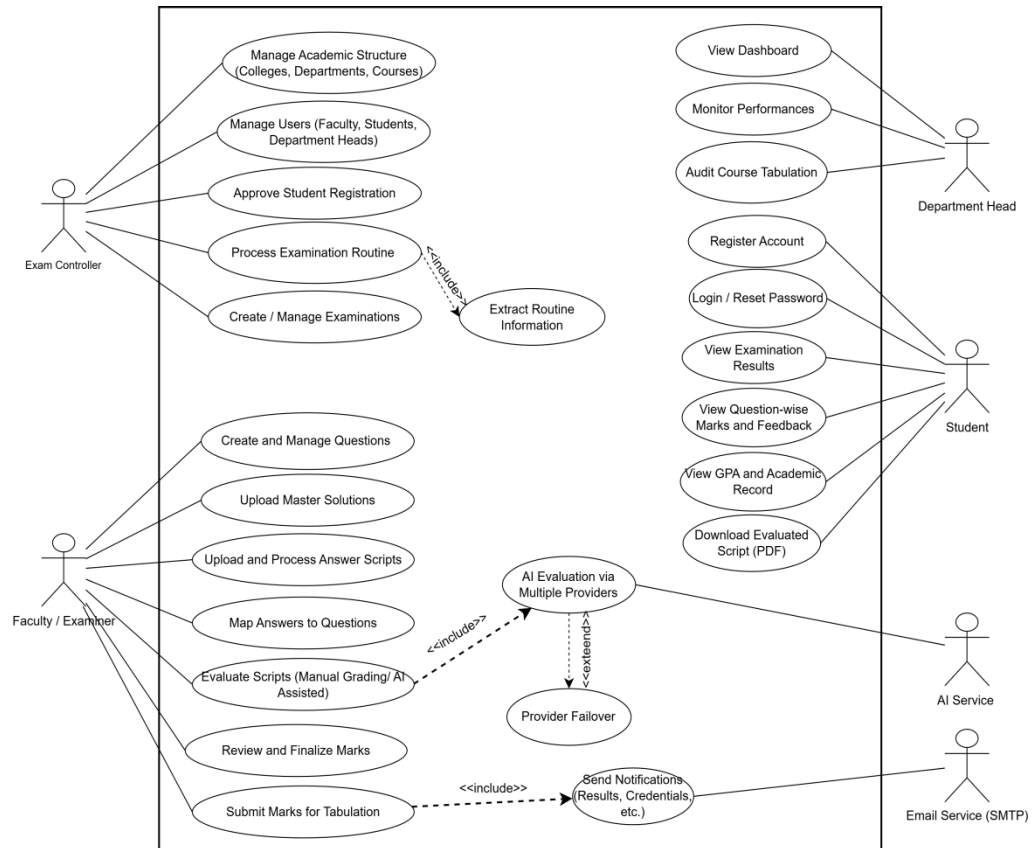


Figure 3.1 Use Case Diagram of IntelliGrade

Chapter 4. Analysis Modeling

4.1 Activity Diagrams

Activity diagrams capture the flow of key processes within IntelliGrade. Seven major workflows are documented below.

4.1.1 Exam Controller Governance Workflow

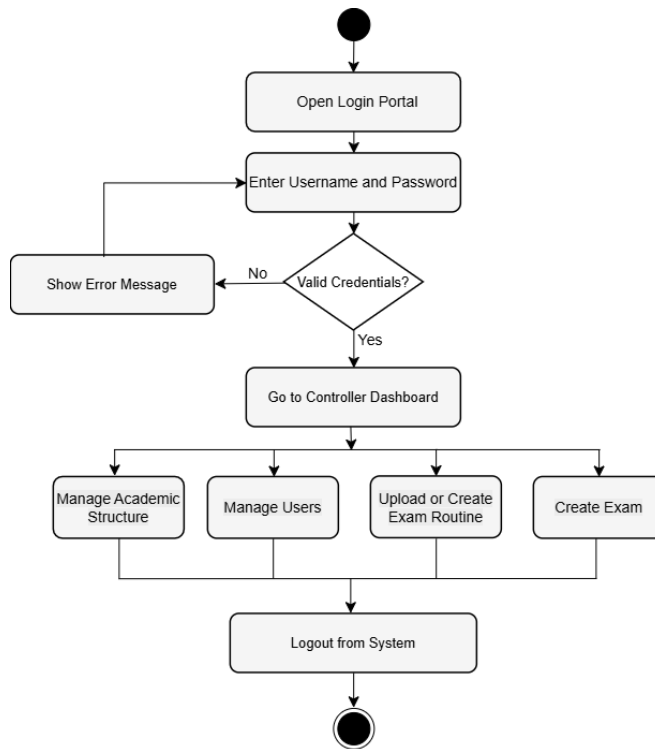


Figure 4.1 Activity Diagram: Exam Controller Governance Workflow

4.1.2 Activity Diagram: Examination Scheduling and Provisioning

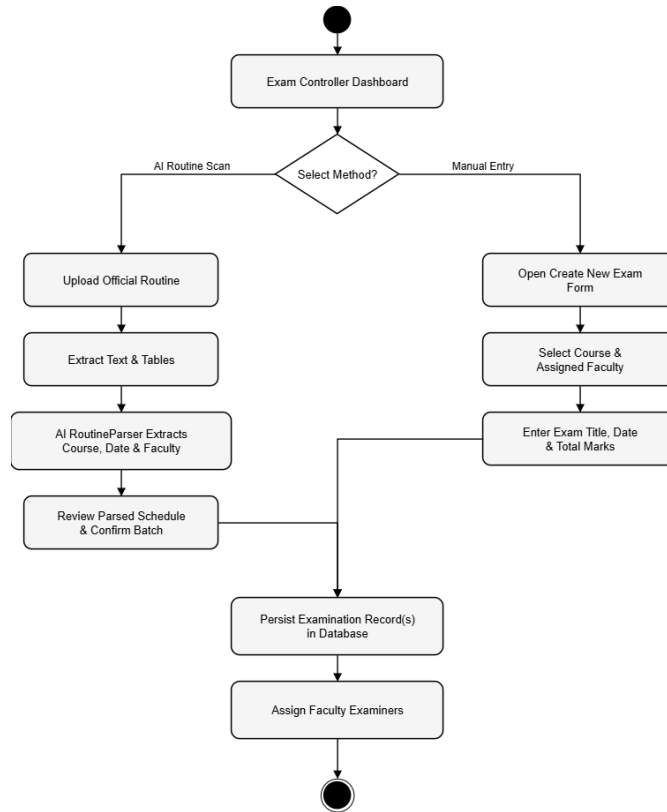


Figure 4.2 Activity Diagram: Examination Scheduling and Provisioning (AI Routine Scan vs. Manual Entry)

4.1.3 Faculty/Examiner Activity Diagram

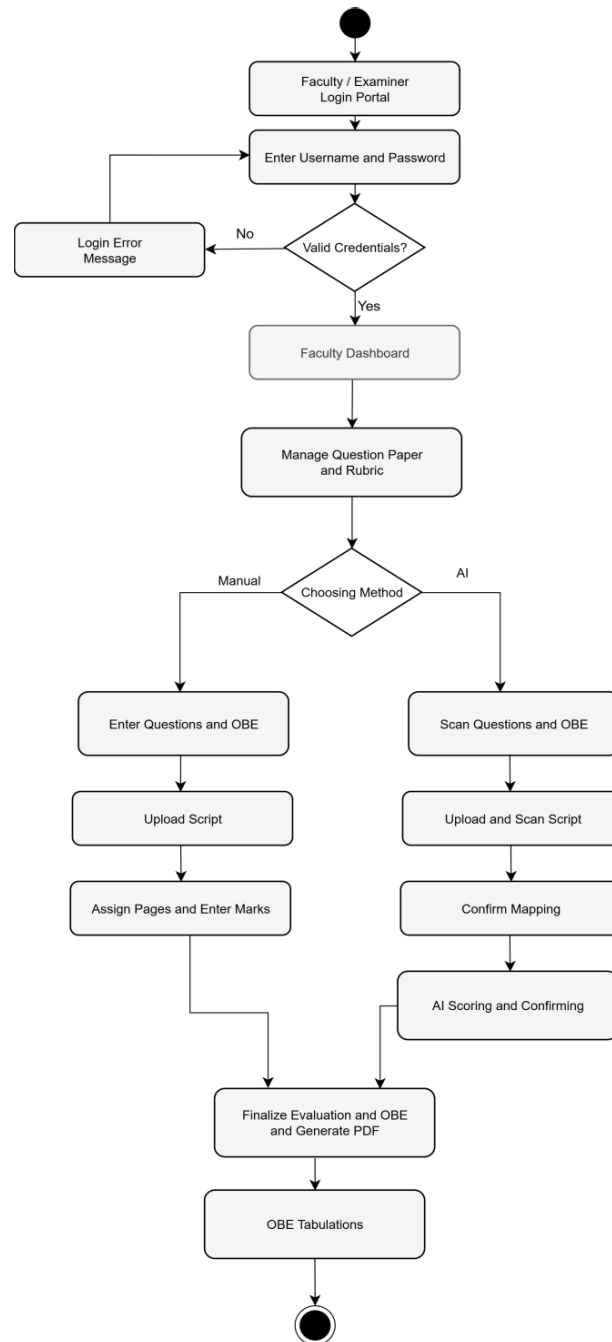


Figure 4.3 Activity Diagram: Faculty/Examiner Workflow

Faculty members navigate to the Question Paper Studio for a specific examination. They can author questions manually, providing the question prompt, maximum marks, and all 23 OBE taxonomy fields for each question, or they can upload the official question paper PDF and trigger AI-assisted scanning that automatically extracts question numbers, marks allocations, and suggested taxonomy tags. For each question, a structured rubric is defined including grading criteria, ideal answer, mark distribution across sub-criteria, expected keywords, and common student mistakes.

4.1.4 Question Paper and Rubric Authoring

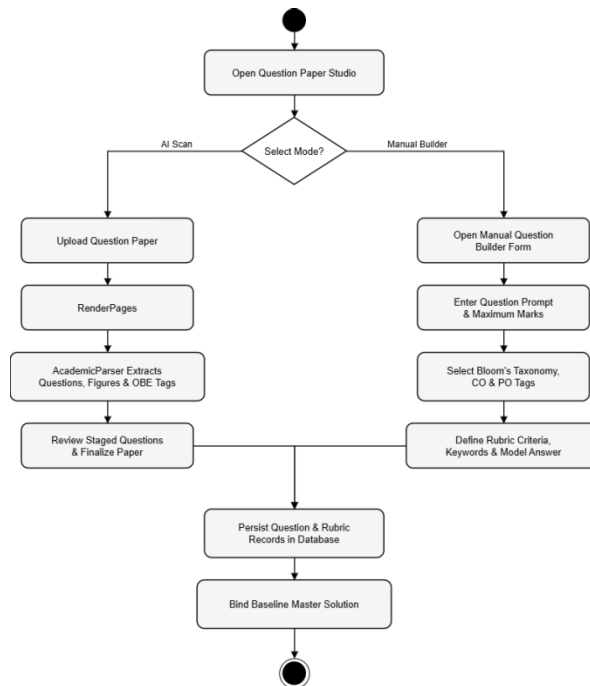


Figure 4.4 Activity Diagram: Question Paper and Rubric Authoring (AI Document Scan vs. Manual Builder)

4.1.5 AI Evaluation Wizard Flow

The AI Evaluation Wizard provides an assistive grading pathway designed to accelerate script marking while maintaining teacher oversight. The workflow executes an end-to-end cascade: preprocessing high-resolution 300 DPI answer script images, performing hybrid OCR extraction, running rubric-aligned inference through a multi-provider failover chain (local Moondream2, Groq, OpenRouter, Gemini, and OpenAI), and populating suggested marks and pedagogical justifications directly into the teacher's verification queue. Crucially, marks generated by the AI wizard remain non-binding drafts until explicitly reviewed, modified, or approved by the course instructor.

Figure 4.5 Activity Diagram: AI Evaluation Wizard Flow

4.1.6 Manual Script Grading Wizard Flow

The Manual Script Grading Wizard operates as an independent, high-speed grading interface for faculty members who prefer direct evaluation or when cloud AI services are unavailable.

The instructor views the scanned script with zoom and pan controls, references the question-specific OBE rubric criteria alongside Bloom's taxonomy levels, inputs scores question-by-question with immediate validation, and commits the final grade. This guarantees that IntelliGrade remains 100% operational as a self-sufficient institutional examination management platform regardless of network connectivity or AI provider rate limits.

Figure 4.6 Activity Diagram: Manual Script Grading Wizard Flow

4.1.7 OBE Tabulation

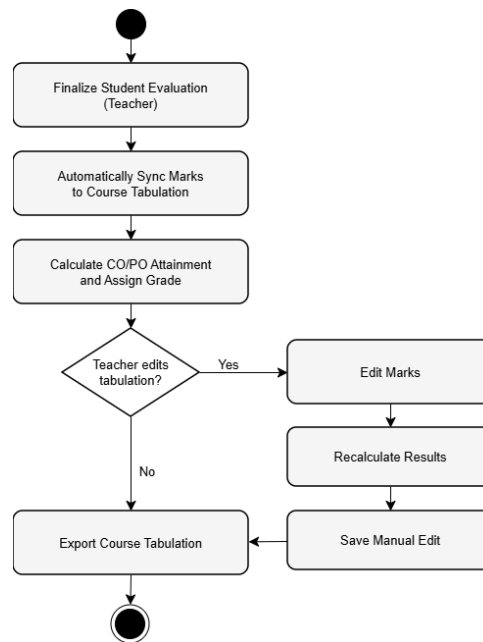


Figure 4.7 Activity Diagram: OBE Tabulation & Calculation

4.1.8 Department Head

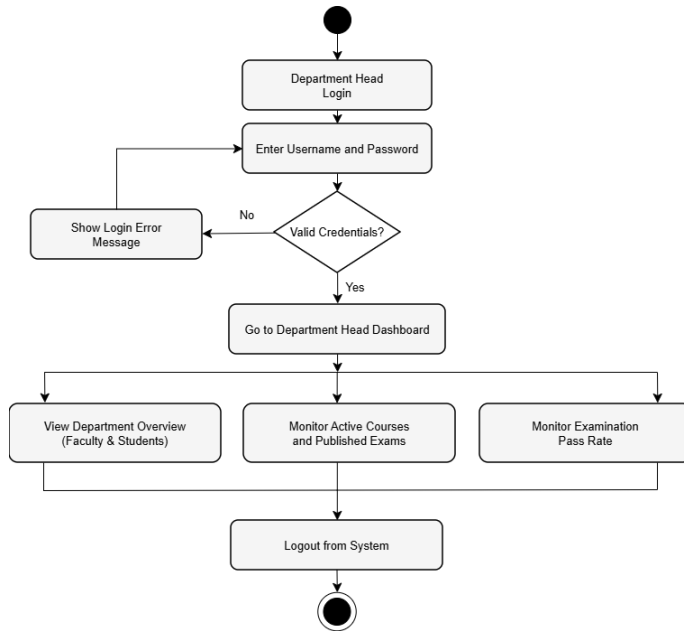


Figure 4.8 Activity Diagram: Department Head Governance

4.1.9 Student Portal

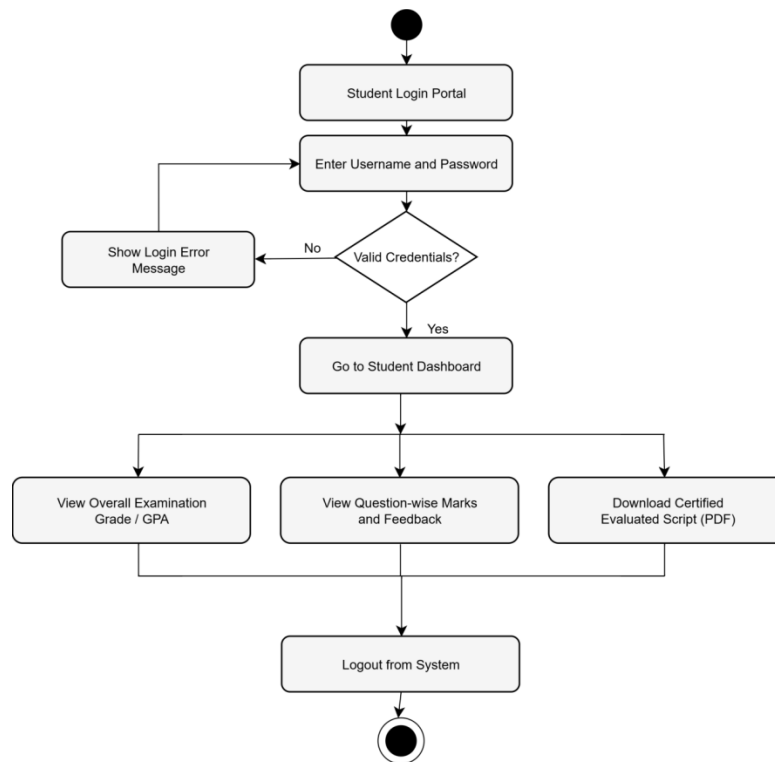


Figure 4.9 Activity Diagram: Student Academic Transparency Portal

4.2 Swim-lane Diagram

4.2.1 Swim-lane Diagram for User Authentication & Role-Based Access Control

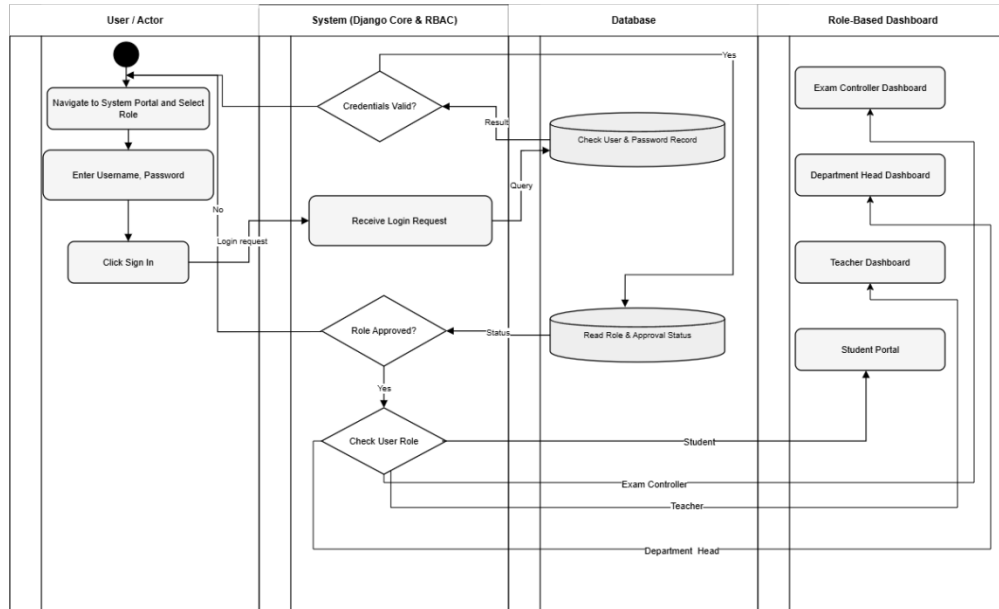


Figure 4.10 Swim-lane Diagram for User Authentication & Role-Based Access Control

4.2.2 Swim-lane Diagram for Examination Scheduling & Faculty Allocation

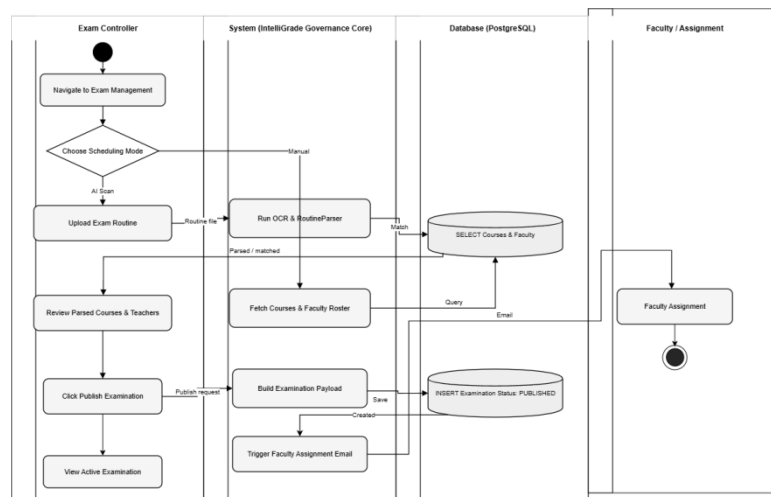


Figure 4.11 Swim-lane Diagram for Examination Scheduling & Faculty Allocation

4.2.3 Swim-lane Diagram for Answer Script Evaluation & Finalization

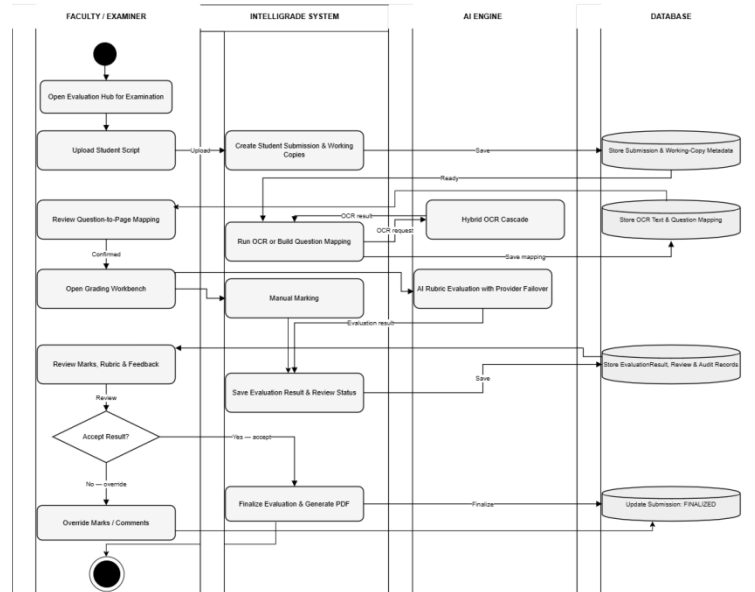


Figure 4.12 Swim-lane Diagram for Answer Script Evaluation & Teacher Finalization

4.2.4 Swim-lane Diagram for OBE Tabulation, Excel Export & Student Result Dissemination

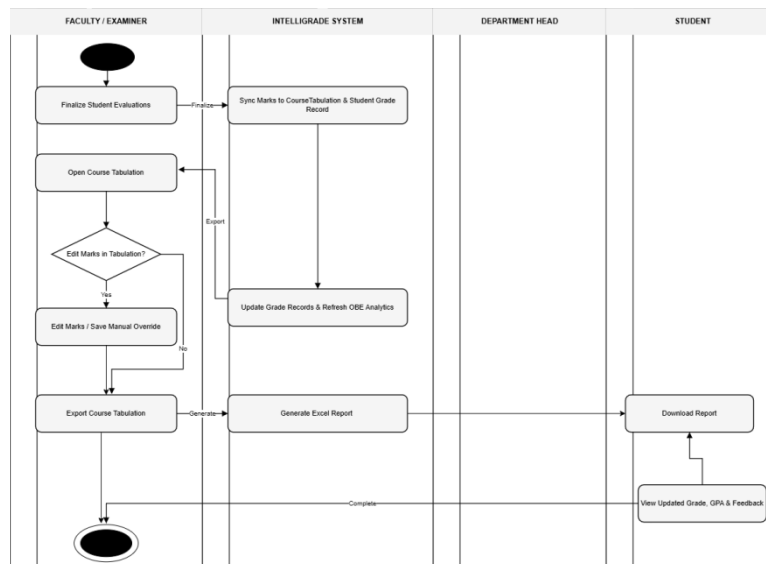


Figure 4.13 Swim-lane Diagram for OBE Tabulation, Excel Export & Student Result Dissemination

4.4 CRC Cards

Class-Responsibility-Collaborator (CRC) cards document the primary classes in IntelliGrade, their responsibilities, and their collaborators.

Table 1 CRC Card: Profile (User Role)

Class Name: Profile	
Class Type: Role	
Class Characteristics: Tangible, Persistent, Sequential	
Responsibilities • Store role (ADMIN, TEACHER, STUDENT, DEPT_HEAD) • Store approval status (is_approved) • Enforce role-based portal access • Store OTP reset codes with expiry	Collaborators • User (Django Auth) • Department • Examination

Table 2 CRC Card: Examination

Class Name: Examination	
Class Type: Entity	
Class Characteristics: Tangible, Persistent, Sequential	
Responsibilities • Store exam type (MID/FINAL/CT/QUIZ), status, and dates • Store approval status (is_approved)	Collaborators • Course • Question • StudentSubmission

• Hold uploaded question paper, rubric, and master solution files • Manage publication workflow (DRAFT → PUBLISHED → CLOSED) • Expose has_master_solution property	• Profile (assigned faculty)
--	--

Table 3 CRC Card: Question

Class Name: Question	
Class Type: Entity	
Class Characteristics: Tangible, Persistent, Sequential	
Responsibilities • Store 23-taxonomy OBE fields (Bloom, CO, PO, KP, CEP, CEA)• Hold uploaded question paper, rubric, and master solution files • Store prompt text and maximum marks• Expose has_master_solution property • Reference attached figures, tables, and formula objects • Provide formatted_number property for display	Collaborators • Examination • Rubric (OneToOne) • QuestionFigure, QuestionTable, QuestionFormula • SubmissionAnswer

Table 4 CRC Card: StudentSubmission

Class Name: StudentSubmission	
Class Type: Entity	
Class Characteristics: Tangible, Persistent, Sequential	
Responsibilities • Manage 14-state FSM lifecycle • Store uploaded script file and extracted OCR data • Track finalization status (is_finalized) • Maintain composite B-tree database indexes • Record student identity (name, roll number)	Collaborators • Examination • SubmissionPage, SubmissionImage • QuestionMapping • EvaluationAuditLog

Table 5 CRC Card: EvaluationResult

Class Name: EvaluationResult	
Class Type: Entity	
Class Characteristics: Tangible, Persistent, Sequential	
Responsibilities • Store obtained marks, maximum marks, and confidence score • Store AI-generated feedback, strengths, and mistakes as JSON	Collaborators • SubmissionAnswer (OneToOne) • TeacherReview • EvaluationHistory • EvaluationFeedback

• Flag low-confidence answers for mandatory teacher review • Manage review status (PENDING/APPROVED/OVERRIDDEN/REJECTED)	
---	--

Table 6 CRC Card: CourseTabulation

Class Name: CourseTabulation	
Class Type: Entity	
Class Characteristics: Tangible, Persistent, Sequential	
Responsibilities • Store semester, section, and weightage configuration • Maintain uniqueness constraint (course, semester, section) • Provide aggregated assessment weightage configuration as JSON	Collaborators • Course • StudentGradeRecord

Table 7 CRC Card: StudentGradeRecord

Class Name: StudentGradeRecord
Class Type: Entity
Class Characteristics: Tangible, Persistent, Sequential

Responsibilities	Collaborators
• Store exam score breakdown per category (CT, Mid, Final, etc.) • Compute weighted overall score via <code>get_category_data()</code> • Store CO/PO attainment scores as JSON vectors • Expose typed properties: <code>class_test_data</code>, <code>midterm_data</code>, <code>final_data</code> • Track manual instructor overrides (<code>is_manually_edited</code>)	• CourseTabulation

4.5 Class Diagram

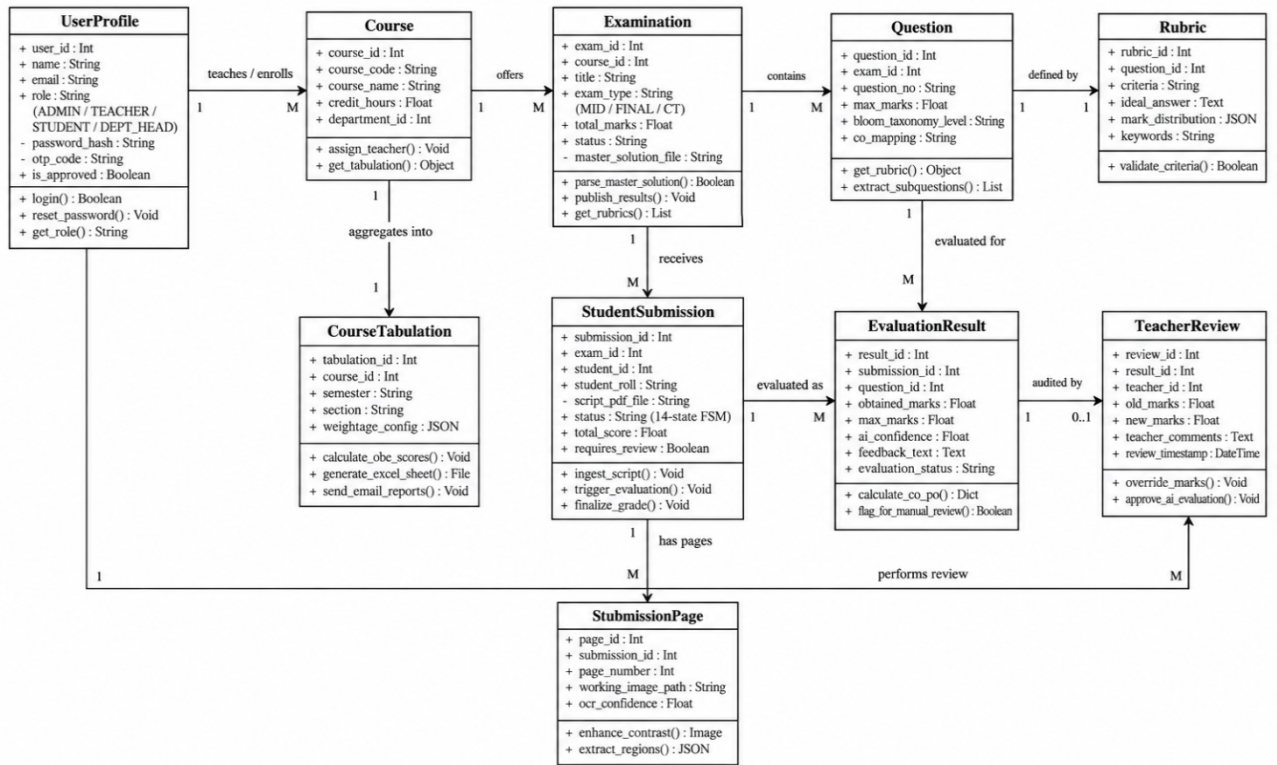


Figure 4.14 Class Diagram: Core Domain Models

Chapter 5. Project Management

5.1 Risk Identification

Table 8 Risk Identification

Risk ID	Risk Description	Category	Probability	Impact
R-01	CPU-bound EasyOCR latency renders AI evaluation too slow for practical use on large exam batches	Technical	High	High
R-02	Cloud AI provider rate limits (HTTP 429) cause evaluation stalls and timeouts during concurrent batch grading	Technical	High	High
R-03	Extreme handwriting illegibility causes OCR failure and unmappable question boundaries	Technical	Medium	High
R-04	Data loss or corruption during evaluation due to unhandled server exceptions	Technical	Low	Critical
R-05	Institutional SMTP credential failure causes email notification breakdown	Operational	Medium	Medium
R-06	Growing media file storage causes disk space overflow on the production server	Operational	Medium	High
R-07	OBE calculation discrepancies due to floating-point rounding in Python arithmetic	Technical	Low	High

5.2 Risk Analysis

Table 9 Risk Analysis

Risk ID	Risk Description	Likelihood (1-5)	Severity (1-5)	Risk Exposure Score	Priority
R-01	CPU OCR latency (EasyOCR ~20-25s/page on CPU)	5	4	20	Critical
R-02	AI rate limiting — HTTP 429 cascade stalls	5	4	20	Critical
R-03	Handwriting illegibility → OCR failure	3	4	12	High
R-04	Data loss during evaluation exception	2	5	10	High
R-05	SMTP email delivery failure	3	3	9	Medium
R-06	Disk storage overflow from working images	3	4	12	High
R-07	Floating-point rounding in OBE calculations	2	4	8	Medium

5.3 Risk Mitigation, Monitoring and Management (RMMM)

Table 10 RMMM: Risk 1 — CPU OCR Latency

Field	Details
Risk ID	R-01
Risk Description	CPU-bound EasyOCR CRAFT+BiLSTM deep learning model takes approximately 20-25 seconds per script page on a modern CPU, making full 13-page script evaluation take over 4 minutes — impractical for end-of-semester batch grading.
Mitigation	1. Apply 800px LANCZOS downsampling before local vision dispatch (implemented). 2. Provide

	Manual Grading Wizard as a zero-OCR fallback for urgent evaluations. 3. Plan GPU CUDA upgrade path (NVIDIA RTX 4060, 8GB VRAM) in hardware budget.
Monitoring	AIProviderHealth model tracks avg_response_time_ms. Dashboard alerts configured for latencies exceeding 60 seconds per page.
Contingency	Fall back to Manual Wizard for urgent evaluations while GPU upgrade is provisioned. Teachers can still evaluate all students; only the automated text extraction is slowed.
Owner	System Architect
Review Date	End of each examination period

Table 11 RMMM: Risk 2 — AI Rate Limiting

Field	Details
Risk ID	R-02
Risk Description	Free-tier cloud AI APIs impose strict rate limits. Groq enforces 30 RPM / 6,000 TPM, Gemini enforces 15 RPM / 1,000,000 TPM. Evaluating multiple questions simultaneously for multiple students triggers HTTP 429 quota exhaustion.
Mitigation	1. ProviderHealthTracker with 120-second cooldown registry per provider (implemented). 2. Maintain 5-provider failover chain so no single HTTP 429 blocks evaluation. 3. Apply JSON sanitization and LaTeX bracket repair to prevent re-requests due to parse errors.
Monitoring	AIProviderHealth.error_count and cooldown_until tracked in real time. Dashboard badge alerts teacher when all providers are rate-limited simultaneously.
Contingency	Upgrade to enterprise Pay-As-You-Go API tiers (Groq Developer Plan, Gemini Tier 1) for production deployment. Fall back to manual workbench if all AI providers are exhausted.
Owner	Exam Controller
Review Date	Before each examination period

Table 12 RMMM: Risk 3 — Handwriting Illegibility

Field	Details
Risk ID	R-03
Risk Description	Some students produce extremely poor handwriting that defeats even EasyOCR's deep learning recognizer, resulting in empty or garbled OCR output and unmapped question boundaries.

Mitigation	1. Three-engine OCR cascade (PyMuPDF → PyTesseract → EasyOCR) maximizes text recovery probability. 2. Interactive mapping modal allows manual page assignment when auto-detection fails. 3. For illegible scripts, vision-capable AI providers (Gemini 2.5 Flash, GPT-4o) evaluate directly from the page image, bypassing OCR text entirely.
Monitoring	OCR confidence scores stored per page in SubmissionPage.ocr_confidence. Pages below threshold flagged in UI for teacher attention.
Contingency	Teacher manually selects pages via the visual confirmation modal, which is always available regardless of OCR output quality.
Owner	Faculty Member
Review Date	Per evaluation session

Table 13 RMMM: Risk 4 — Data Loss During Evaluation

Field	Details
Risk ID	R-04
Risk Description	An unhandled server exception or network interruption mid-evaluation could corrupt the StudentSubmission state machine or lose partially written SubmissionAnswer records.
Mitigation	1. All database writes wrapped in django.db.transaction.atomic() ensuring partial states are rolled back on failure. 2. EvaluationAuditLog records every state transition with timestamp and actor identity. 3. Script PDFs and working images stored on disk and only purged after explicit teacher-initiated finalization. 4. extracted_ocr_data JSON field caches OCR output so re-evaluation does not require re-processing all images.
Monitoring	EvaluationAuditLog monitored for FAILED state transitions. System error log reviewed after each evaluation session.
Contingency	Re-upload script and restart evaluation pipeline from UPLOADED state. All previously extracted OCR data is available for reference.
Owner	System / Admin
Review Date	Continuous

Table 14 RMMM: Risk 5 — SMTP Email Delivery Failure

Field	Details
Risk ID	R-05
Risk Description	SMTP server outages or expired credentials can cause failure in account creation emails, OTP

	password reset delivery, and graded result notification emails.
Mitigation	1. Email dispatch runs in a background threading.Thread so SMTP latency or failure never blocks the HTTP response seen by the user. 2. Email failures are logged to the application error log without raising user-visible exceptions. 3. OTP reset includes fallback instructions for Exam Controller manual intervention.
Monitoring	SMTP error logs reviewed daily during active grading periods.
Contingency	Exam Controller can manually reset passwords via Django admin. Results can be communicated as PDF downloads shared manually.
Owner	Exam Controller
Review Date	Monthly

Table 15 RMMM: Risk 6 — Disk Storage Overflow

Field	Details
Risk ID	R-06
Risk Description	300 DPI page images (~26 MB uncompressed per page) accumulate rapidly. A course with 60 students x 13 pages generates approximately 20 GB of working copies, which can exhaust institutional server disk space.
Mitigation	1. FinalizationService._purge_temporary_artifacts() deletes submission_working/ and submission_preview/ upon certified PDF creation (implemented). 2. JPEG quality=75 compression applied to all working copies. 3. 800px LANCZOS downsampling for local vision payloads reduces per-image size by ~85%.
Monitoring	Server disk usage alerts configured at 80% threshold. Pending finalization queue monitored via admin dashboard to identify abandoned submissions.
Contingency	Emergency purge script available to delete working images for all finalized submissions. Future plan: migrate to AWS S3 / Cloudflare R2 object storage with automatic lifecycle expiration policies.
Owner	System Admin
Review Date	End of each examination period

Table 16 Risk 7 — OBE Calculation Inconsistency

Field	Details
Risk ID	R-07

Risk Description	Python floating-point arithmetic without explicit rounding can produce weighted OBE scores that differ by ± 0.01 from expected values, causing discrepancies between the web tabulation view and the official exported Excel workbook.
Mitigation	1. All intermediate scores wrapped in <code>round(value, 2)</code> at each aggregation step in <code>get_category_data()</code> . 2. Generated Excel outputs use <code>openpyxl</code> 's native formula support for in-sheet validation against Python-computed values. 3. Automated test suite validates OBE calculation correctness against 50 manually verified student records.
Monitoring	Automated test suite (<code>tests.py</code> , 1000+ assertions) run after every code change. OBE output spot-checked by faculty against manual calculations each semester.
Contingency	Teacher can manually override any computed score via the tabulation modal, with override flagged in <code>is_manually_edited</code> for auditability.
Owner	Developer
Review Date	After every deployment

Chapter 6. Project Planning and Scheduling

6.1 Function Point Analysis

Function Point Analysis (FPA) was used to estimate the size and complexity of IntelliGrade's software, providing an objective basis for effort and schedule estimation independent of programming language or implementation technology.

Table 17 Functionality, Input and Output

Function	Function Type	Description	Input	Output
User Authentication and RBAC	ILF	Profile, User, session management	Username / Password	Authenticated session + role redirect
AI Routine Scanner	EI	Parse multi-page exam schedule PDFs	PDF / Image file	Structured Examination records
23-Taxonomy Question Studio	ILF	Store questions with 23 OBE fields	Question form fields	Saved Question + Rubric records
Script Preprocessing	EI	Rasterize PDFs, skew, threshold	PDF / JPG / ZIP	300 DPI working copy images
Hybrid OCR Pipeline	EO	Multi-engine text extraction	Page images	Text + bounding box JSON
Question Boundary Detection	ILF	Regex state machine heading detection	OCR text per page	QuestionMapping records
AI Evaluation Failover Engine	EIF	Multi-provider AI orchestration	Prompt + context	EvaluationResult JSON
Split-Screen Workbench	EO	Interactive grading interface	Submission ID	Mark overrides + audit log entries
Certified PDF Generation	EO	ReportLab PDF stamping	Finalized submission	Watermarked evaluated PDF file
OBE Tabulation Engine	ILF	Weighted score aggregation	Exam scores JSON	StudentGradeRecord with CO/PO
8-Sheet Excel Export	EO	openpyxl workbook generation	Grade records	XLSX workbook (8 sheets)
Asynchronous Email Service	EIF	SMTP email dispatch	Event trigger + payload	Delivered email with PDF attachment
Student Dashboard	EO	Transparency grade portal	Student session	Live grade cards + download links

Table 18 Complexity of Data Function

Data Function	Elements	Records	Complexity
Profile / User	8	4 roles	Simple
Examination + Question + Rubric	25	Nested hierarchy	Complex
StudentSubmission + Pages	18	Per exam per student	Average
EvaluationResult + Feedback	15	Per answer per question	Average
CourseTabulation + GradeRecord	20	Per course per semester	Complex
AIConfiguration + ProviderHealth	12	Global singleton	Simple

Table 19 Complexity of Transaction Function

Transaction Function	Type	Complexity
User Login and Role Redirect	EI	Simple
AI Routine Scanning	EI	Complex
Question Paper Authoring (23-field taxonomy)	EI	Complex
Script Upload + 300 DPI Preprocessing	EI	Complex
Hybrid OCR Cascade	EO	Complex
AI Evaluation with Multi-Provider Failover	EIF	Complex
Teacher Mark Override + Audit Logging	EI	Average
Finalize + Certified PDF + Storage Purge	EO	Complex
OBE Tabulation Aggregation	ILF	Complex
8-Sheet Excel Export	EO	Complex
Asynchronous SMTP Email	EIF	Average
Student Grade Dashboard	EO	Average

Table 20 UFP Calculation Summary

Function Point Category	Count	Weight	Contribution
External Inputs (EI)	6	4	24
External Outputs (EO)	7	5	35
External Interface Files (EIF)	3	7	21
Internal Logical Files (ILF)	5	10	50

Total Unadjusted Function Points (UFP)			130
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Table 21 Technical Difficulty Index (TDI)

Technical Factor	Rating (0-5)	Description
Data Communications	5	Multi-provider REST APIs, AJAX, SMTP
Distributed Processing	4	Background email threads, async evaluation
Performance	5	Sub-second DB queries, 8-sec AI target
Complex Processing	5	OCR cascade, AI failover, PDF stamping
Reusability	3	Modular AI engine provider classes
Ease of Installation	2	Docker + env configuration required
Ease of Operation	4	Web-based, role-guided UI
Portability	3	Django + PostgreSQL portable
Maintainability	4	Modular architecture, clean separation
TDI Total	40	$VAF = 0.65 + (0.01 \times 40) = 1.05$

$$\text{Adjusted Function Points (AFP)} = \text{UFP} \times \text{VAF} = 130 \times 1.05 = 136.5 \text{ AFP}$$

At a productivity rate of approximately 8 AFP per person-month for a complex real-time system with AI integration and handwriting recognition, the estimated development effort is approximately 17 person-months, which is consistent with the actual observed development timeline of 8 calendar months with a 5-person team equivalent.

6.2 Project Schedule (Gantt Chart)

Task	Subtask	Month 1 (July)				Month 2 (August)				Month 3 (September)			
		W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Planning & Requirement Gathering	Background Study & Literature Review												
	Requirement Specifications & SRS												
Analysis & Modeling	Use Case & User Stories												
	Activity & Sequence Diagrams												
Design	DB Schema & 14-State FSM Design												
	UI Wireframes & Architecture Design												
Coding & Implementation	Core Engine (OCR, AI Failover & Rubrics)												
	OBE Tabulation & Role Dashboards												
Testing & Finalization	E2E Unit & Integration Testing												
	Audit, Documentation & Final Review												

Figure 6.1 Gantt Chart: Project Schedule

Chapter 7. Cost Estimation

7.1 Personnel Cost

Table 22 Personnel Salary

Role	No. of People	Monthly Salary (BDT)
System Architect / Full-Stack Developer	1	80,000
AI / ML Engineer	1	75,000
Database Engineer	1	60,000
UI/UX Designer	1	50,000
QA / Test Engineer	1	45,000
Total	5	310,000

Table 23 Personnel Cost Estimation

Role	Duration (Months)	Monthly Cost (BDT)	Total (BDT)
System Architect / Full-Stack Developer	8	80,000	640,000
AI / ML Engineer	6	75,000	450,000
Database Engineer	4	60,000	240,000
UI/UX Designer	4	50,000	200,000
QA / Test Engineer	3	45,000	135,000
Total Personnel Cost			1,665,000

7.2 Hardware Cost

Table 24 Hardware Cost

Item	Specification	Cost (BDT)
Development Workstation	Intel Core i7, 32GB RAM, 1TB SSD	120,000
GPU for Local OCR (Optional)	NVIDIA RTX 4060 (8GB VRAM)	65,000
Network Equipment	Router, switches, UPS	25,000
Total Hardware Cost		210,000

7.3 Software Cost

Table 25 Software Cost

Item	Description	Annual Cost (BDT)
Python / Django / PostgreSQL	Open-source — Free	0
EasyOCR / PyTesseract / PyMuPDF / OpenCV	Open-source — Free	0
Groq Cloud API (Free Tier)	30 RPM / 6,000 TPM	0
Gemini API (Free Tier)	15 RPM / 1,000,000 TPM	0
PythonAnywhere Hosting (Hacker Plan)	Production deployment	6,720
Domain Name Registration	.com domain	1,500
Total Software Cost		8,220

7.4 Operational Cost

Table 26 Other Operational Costs

Item	Description	Monthly Cost (BDT)
Electricity and Internet	Development environment utilities	5,000
Cloud Storage (Future)	AWS S3 or Cloudflare R2 for script archival	3,000
Miscellaneous	Printing, testing materials, etc.	2,000
Total Monthly Operational		10,000

7.5 Total Project Cost Summary

Table 27 Total Project Cost Summary

Category	Cost (BDT)
Personnel Cost (8 months)	1,665,000
Hardware Cost	210,000
Software Cost (Annual)	8,220
Operational Cost (8 months x 10,000/month)	80,000

Grand Total	1,963,220
Grand Total (USD approximate)	~\$17,850

Chapter 8. System Design

8.1 System Architecture Overview

IntelliGrade is a monolithic Django 5.2 web application deployed on a PostgreSQL database backend, organized around a clean service-oriented internal architecture. The application is structured into one primary Django app (core) containing all models, views, URLs, services, and the AI engine subpackage.

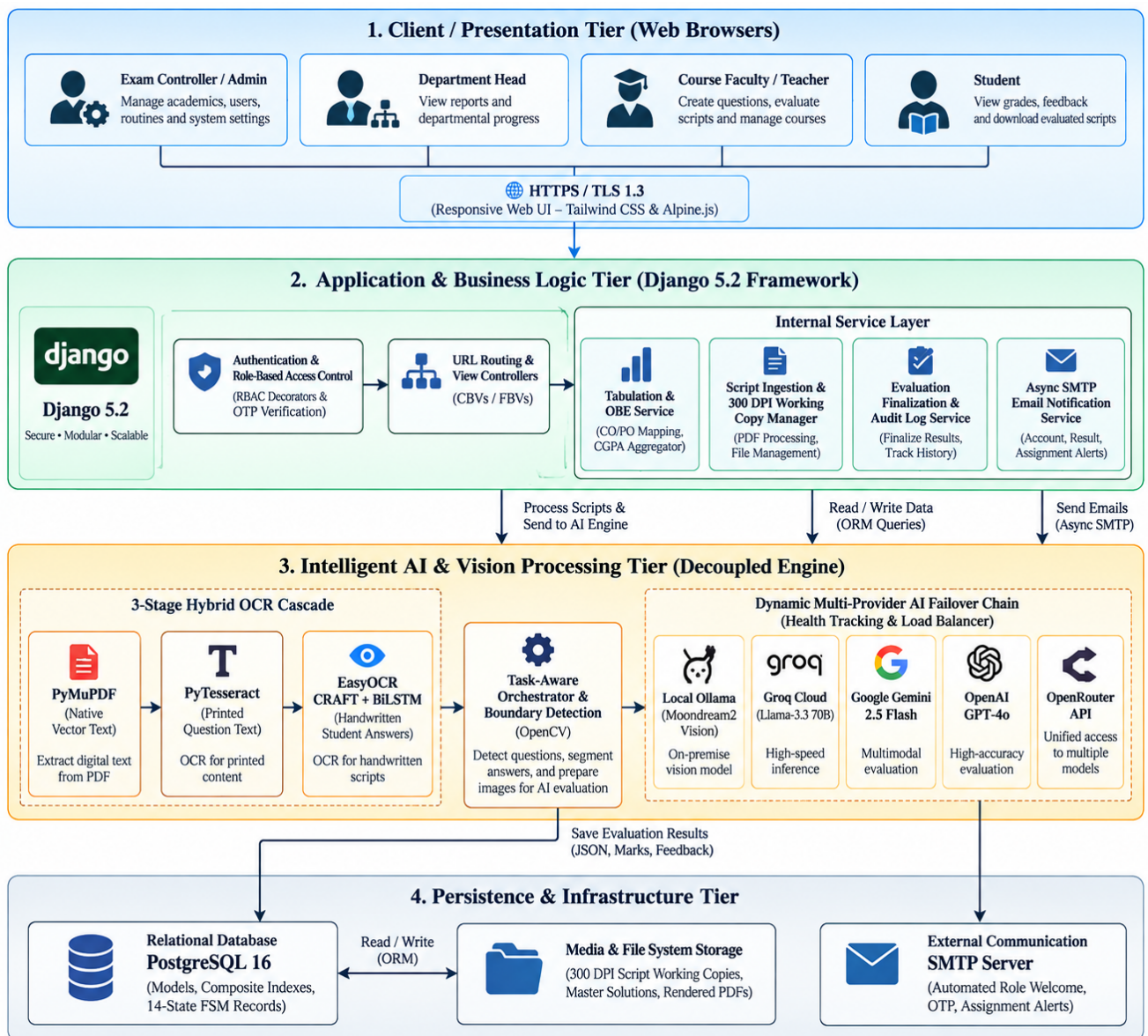


Figure 8.1 System Architecture Overview Diagram (4-Tier)

Table 28 Technology Stack Details

Layer	Technology	Version / Notes
Backend Framework	Django	5.2.x (LTS)
Database	PostgreSQL / SQLite (dev)	SQLite for development, PostgreSQL for production
Computer Vision	OpenCV	4.8.0 (headless)
PDF Rendering	PyMuPDF (fitz)	1.23.0
OCR — Printed Text	PyTesseract	5.3.x
OCR — Handwriting	EasyOCR (CRAFT+BiLSTM)	1.7.0 on PyTorch CPU 2.8.0
AI Provider — Local	Moondream2 via Ollama	Offline vision model
AI Provider — Cloud	Groq Llama-3.3 70B	groq>=0.4.0
AI Provider — Cloud	Google Gemini 2.5 Flash	google-generativeai>=0.3.0
AI Provider — Cloud	OpenAI GPT-4o	openai>=1.0.0
AI Provider — Gateway	OpenRouter	REST API via requests
PDF Generation	ReportLab	4.0.0
Excel Export	openpyxl	3.1.0
Image Processing	Pillow	10.0.0
Data Processing	Pandas, NumPy	2.0.0 / 1.24.0
Email	Django SMTP + threading.Thread	Async non-blocking
Deployment	PythonAnywhere / Gunicorn	gunicorn>=21.2.0

8.2 Module Descriptions

IntelliGrade is organized into 21 functional modules:

Module ID	Module Name	Primary Responsibility
MOD-01	Authentication and RBAC	Session auth, role-based routing, OTP reset
MOD-02	Chief Exam Controller Portal	Academic hierarchy CRUD, student approval
MOD-03	Department Head Dashboard	Departmental metrics and OBE audit
MOD-04	Teacher/Examiner Workspace	Exam management and evaluation launch
MOD-05	Student Transparency Portal	Grade cards, GPA, PDF download
MOD-06	Academic Structure Management	College → School → Department → Course hierarchy

MOD-07	AI Exam Routine Parser	Multi-page PDF/image schedule extraction
MOD-08	Question Paper and 23-Taxonomy Studio	OBE-tagged question and rubric authoring
MOD-09	Master Benchmark Solution Service	Golden solution ingestion and step mapping
MOD-10	Script Ingestion and 300 DPI Preprocessor	PDF/image rasterization, deskewing, thresholding
MOD-11	Hybrid Multi-Engine OCR	PyMuPDF → PyTesseract → EasyOCR cascade
MOD-12	Question Boundary and Mapping Engine	Regex state machine + teacher confirmation modal
MOD-13	AI Evaluation Wizard	Full automated pipeline from upload to scored result
MOD-14	Manual Script Grading Wizard	Zero-AI/OCR fast page slicing and assignment
MOD-15	Multi-Provider AI Evaluation Core	5-provider failover with partial credit scoring
MOD-16	TaskRouter and Cooldown Health Tracker	Provider health monitoring and 429 backoff
MOD-17	Split-Screen Teacher Grading Workbench	Side-by-side script viewer and mark override
MOD-18	Certified PDF Script Generator and Cleanup	ReportLab stamping and working image purge
MOD-19	Course OBE Tabulation Engine	Weighted aggregation, letter grades, CO/PO
MOD-20	8-Sheet Excel Export and Bi-directional Sync	openpyxl workbook generation with formulas
MOD-21	Asynchronous Institutional Email Service	Non-blocking SMTP dispatch via background threads

8.3 Data Flow Diagrams (DFD)

8.3.1 DFD Level 0 — Context Diagram

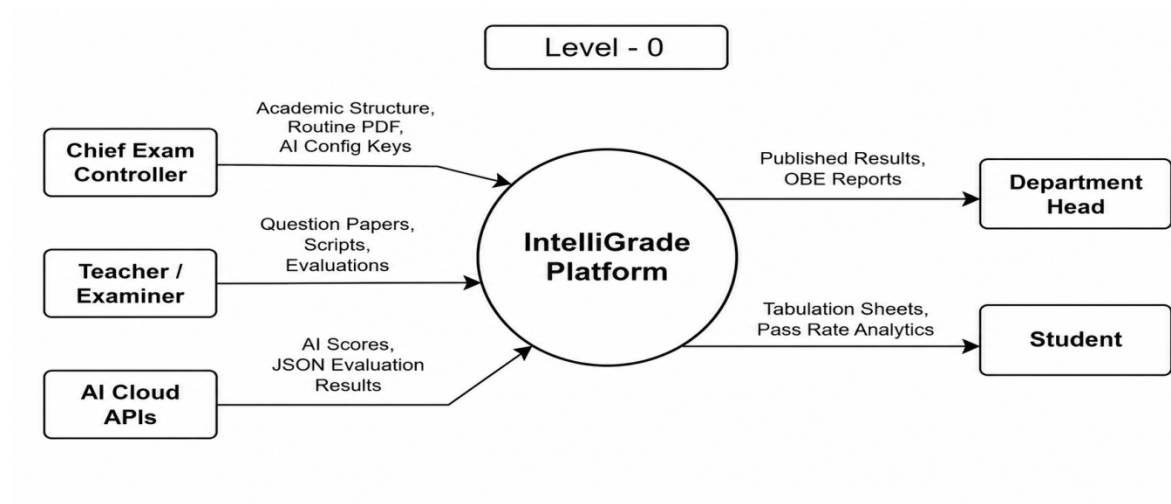


Figure 8.2 Data Flow Diagram — Level 0 (Context Diagram)

VIII.3.2 DFD Level 1

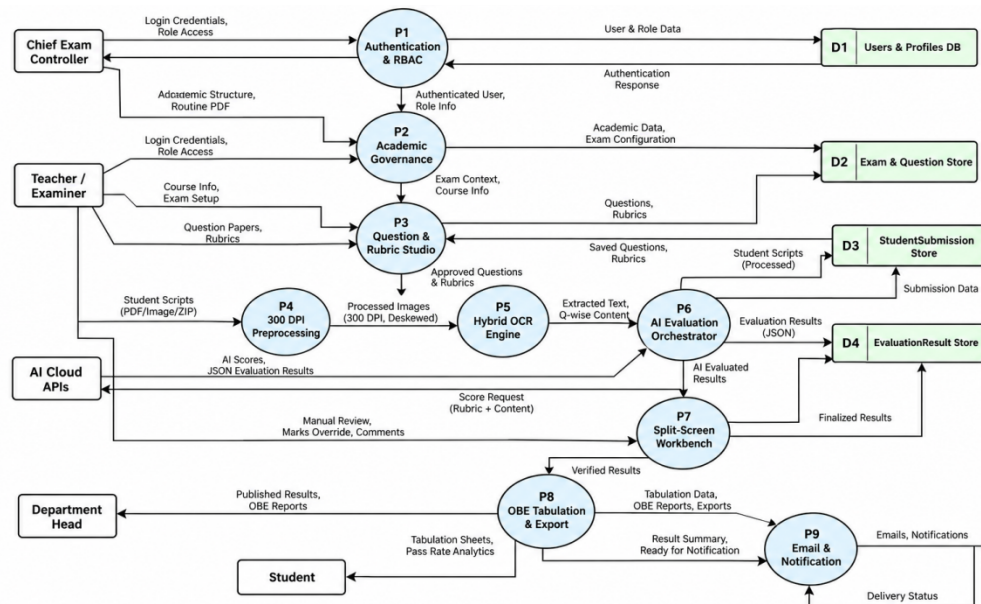


Figure 8.3 Data Flow Diagram — Level 1 (System Level)

VIII.4 Figure 8.4 Entity-Relationship Diagram (ERD)

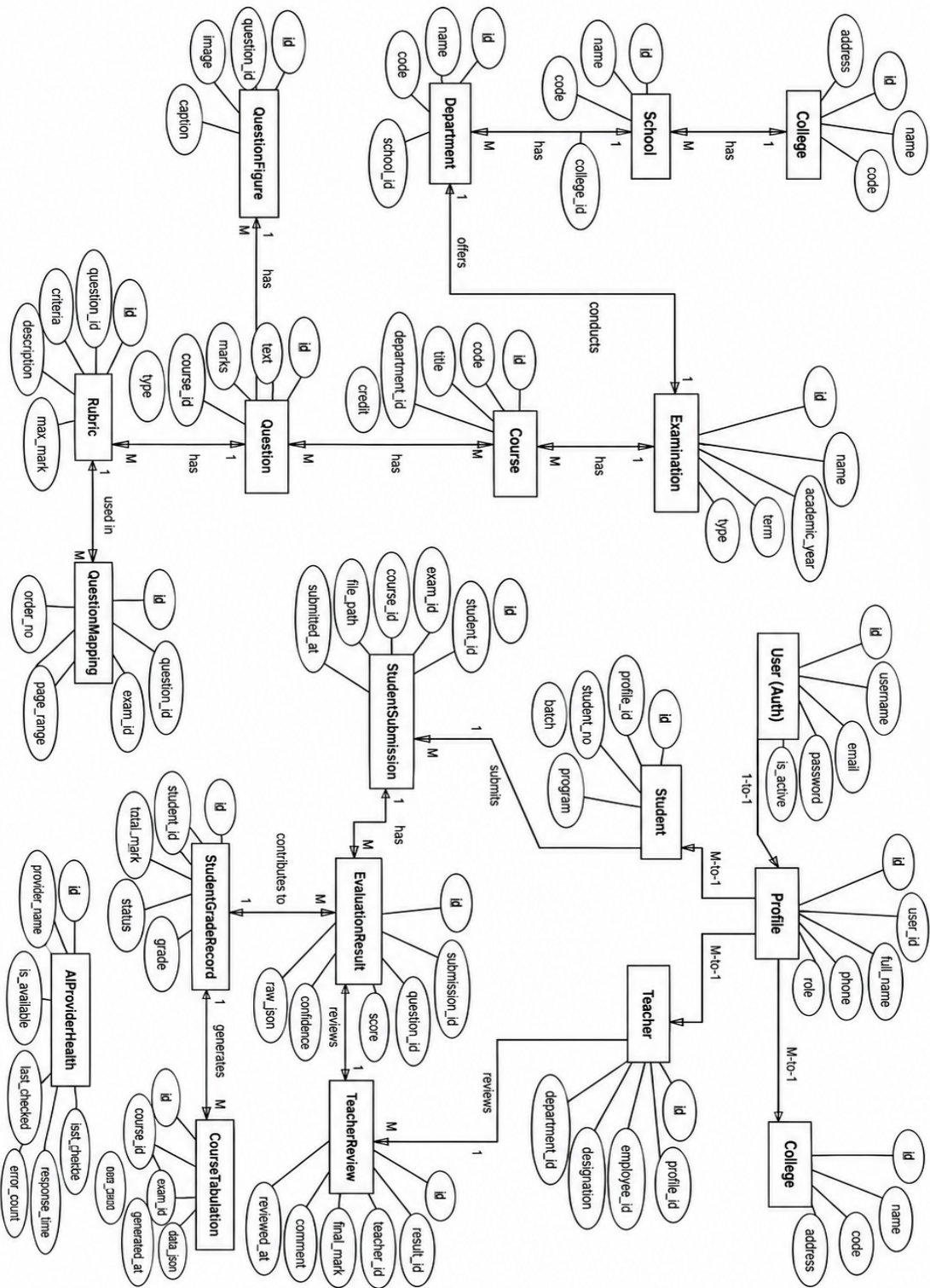


Figure 8.4 Entity-Relationship Diagram (ERD)

8.5 Database Schema

Table 29 Core Database Schema

Model	Primary Fields	Indexes
Profile	user (OneToOne), role (ADMIN/TEACHER/STUDENT/DEPT_HEAD), department (FK), is_approved (Bool)	Index(role, is_approved)
College	name, code (unique)	Index(code)
School	college (FK), name, code	Index(code)
Department	school (FK), college (FK), name, code, is_active	Index(code, is_active)
Course	department (FK), code (unique), title, instructors (M2M)	Index(code, department)
Examination	course (FK), title, exam_type, total_marks, status, exam_date, master_solution_file, master_solution_parsed	Index(course, status)
Question	examination (FK), question_number, prompt_text, max_marks, bloom_level, co_mapping, po_mapping, kp_mapping, cep_mapping, cea_mapping	Index(examination, question_number)
Rubric	question (OneToOne), criteria, ideal_answer, mark_distribution (JSON), keywords, common_mistakes	OneToOne Primary
QuestionFigure	question (FK), page_number, caption, image, bounding_box (JSON), is_master_solution_figure	Index(question)
QuestionTable	question (FK), element_type, caption, cell_json (JSON), rows, columns	Index(question)
QuestionFormula	question (FK), raw_latex, image, is_matrix, bounding_box (JSON)	Index(question)
StudentSubmission	examination (FK), student (FK), student_name, student_roll_no, status (14-state FSM), total_obtained_marks, is_finalized, extracted_ocr_data (JSON)	Index(examination, status), Index(student_roll_no), Index(is_finalized)
SubmissionPage	submission (FK), page_number, page_image, working_image_path, version, ocr_raw_text, ocr_confidence	Index(submission, page_number)
SubmissionImage	submission (FK), original_file, sequence_order, rotation_angle, version	Index(submission, sequence_order)
SubmissionAnswer	submission (FK), question (FK), extracted_text, ocr_confidence, bounding_box_json, page (FK)	Index(submission, question)
EvaluationResult	submission_answer (OneToOne), obtained_marks, maximum_marks, confidence, strengths_json, mistakes_json, rubric_breakdown_json, feedback_text, status, requires_manual_review	Index(status, requires_manual_review)
TeacherReview	evaluation_result (FK), teacher (FK), action, previous_marks, new_marks, review_comments, created_at	Index(evaluation_result, created_at)
EvaluationHistory	evaluation_result (FK), old_marks, new_marks, reason, changed_by (FK), timestamp	Index(evaluation_result)
EvaluationAuditLog	submission (FK), action, actor (FK), old_state, new_state, notes, ip_address, timestamp	Index(submission, timestamp)

QuestionMapping	submission (FK), question (FK), page_numbers_json, confidence, mapping_status, is_confirmed	Index(submission, mapping_status), Index(submission, is_confirmed)
CourseTabulation	course (FK), semester, section, weightage_config (JSON)	UniqueConstraint(course, semester, section)
StudentGradeRecord	tabulation (FK), student_id, student_name, exam_scores (JSON), co_scores (JSON), po_scores (JSON), attendance_marks, overall_score, letter_grade, is_manually_edited	Index(tabulation, student_id), Index(tabulation, overall_score), Index(is_manually_edited)
AIConfiguration	provider, api_key, model_name, confidence_threshold, is_active	Index(provider, is_active)
AIProviderHealth	provider_name, current_model, status, error_count, cooldown_until, avg_response_time_ms, is_healthy	Index(provider_name, is_healthy)

VIII.5 REST and AJAX API Catalog

Table 30 REST and AJAX API Catalog

Method	Endpoint URL	View Function	Auth Role
POST	/api/auth/forgot-password/	api_forgot_password	Public
POST	/api/auth/verify-reset-otp/	api_verify_reset_otp	Public
POST	/api/auth/reset-password/	api_reset_password	Public
GET	/api/courses-and-faculty/	api_get_courses_and_faculty	ADMIN / DEPT_HEAD
POST	/api/publish-exam/	api_publish_exam	ADMIN / TEACHER
POST	/api/scan-question-paper/	api_scan_question_paper	TEACHER / ADMIN
GET	/api/scan-progress/<exam_id>/	api_get_scan_progress	TEACHER / ADMIN
POST	/api/finalize-scanned-paper/	api_finalize_scanned_paper	TEACHER / ADMIN
POST	/api/generate-ai-rubric/	api_generate_ai_rubric	TEACHER / ADMIN
POST	/api/ai-analyze-question-full/	api_ai_analyze_question_full	TEACHER / ADMIN
POST	/api/exam/<exam_id>/upload-master-solution/	api_upload_master_solution	TEACHER
POST	/api/exam/<exam_id>/upload-submission/	upload_student_submission	TEACHER
POST	/api/exam/<exam_id>/upload-raw-images/	api_upload_raw_images	TEACHER

POST	/api/exam/<exam_id>/upload-wizard-pdf/	api_wizard_upload_pdf	TEACHER
GET	/api/submission/<sub_id>/images/	api_get_submission_images	TEACHER
POST	/api/submission/<sub_id>/reorder-pages/	api_reorder_submission_pages	TEACHER
POST	/api/submission/<sub_id>/create-pdf/	api_create_submission_pdf	TEACHER
GET	/api/submission/<sub_id>/progress/	api_get_submission_progress	TEACHER
POST	/api/submission/<sub_id>/run-evaluation-v3/	api_run_evaluation_v3	TEACHER
POST	/api/submission/<sub_id>/analyze-mapping/	api_analyze_question_mapping	TEACHER
POST	/api/submission/<sub_id>/confirm-mapping/	api_confirm_question_mapping	TEACHER
POST	/api/submission/<sub_id>/reevaluate-v3/	api_reevaluate_v3	TEACHER
GET	/api/submission/<sub_id>/download-evaluated-pdf/	api_download_evaluated_pdf	TEACHER / STUDENT
POST	/api/submission/<sub_id>/finalize/	api_finalize_evaluation	TEACHER
POST	/api/submission/<sub_id>/update-info/	api_update_submission_info	TEACHER
POST	/api/submission/<sub_id>/delete/	api_delete_submission	TEACHER
POST	/api/evaluation-result/<res_id>/review/	review_evaluation_answer	TEACHER
POST	/api/tabulation/grade-record/<rec_id>/update/	api_update_student_grade_record	TEACHER
POST	/course/<course_id>/email-tabulation/	email_course_tabulation_report	TEACHER / DEPT_HEAD
GET	/course/<course_id>/export-tabulation/	export_course_tabulation	TEACHER / DEPT_HEAD

8.7 Interface Design

8.7.1 Landig Page

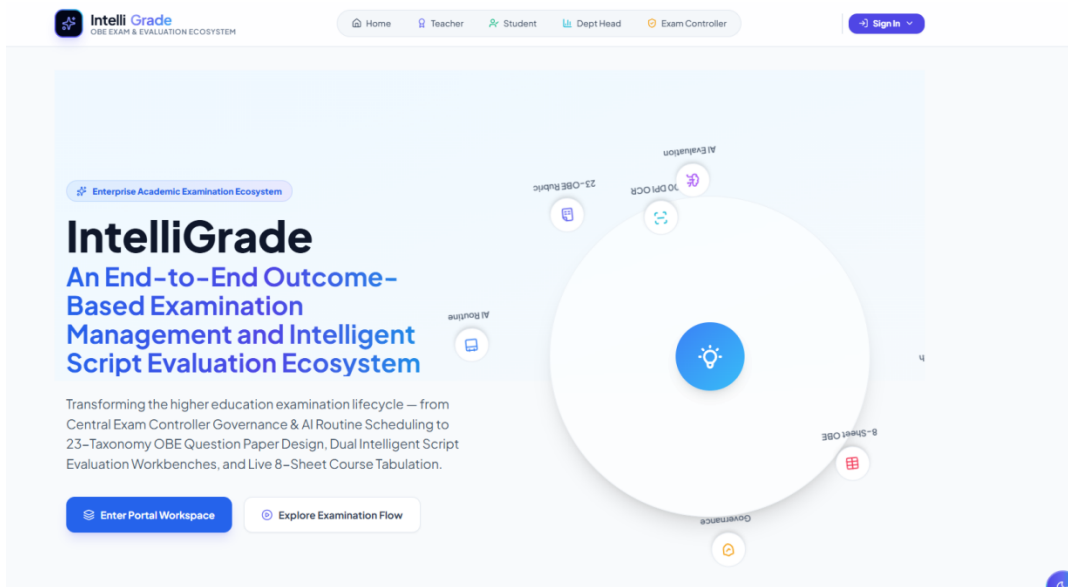


Figure 8.5 UI Mockup: Landing Page & Portal Selector

8.7.2 Exam Controller Dashboard

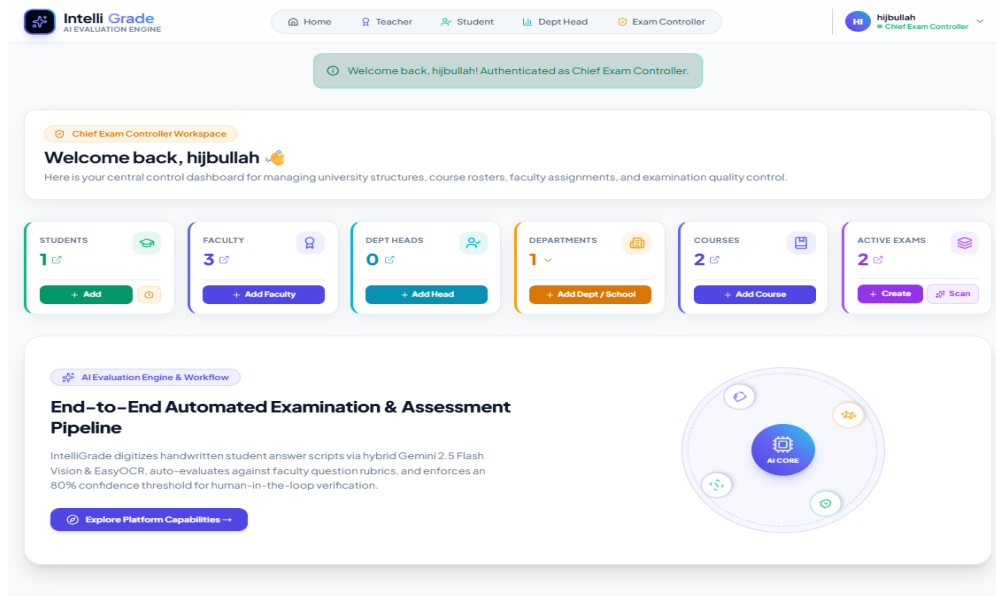


Figure 8.6 UI Mockup: Exam Controller Governance Dashboard

8.7.3 AI Routine Scanner

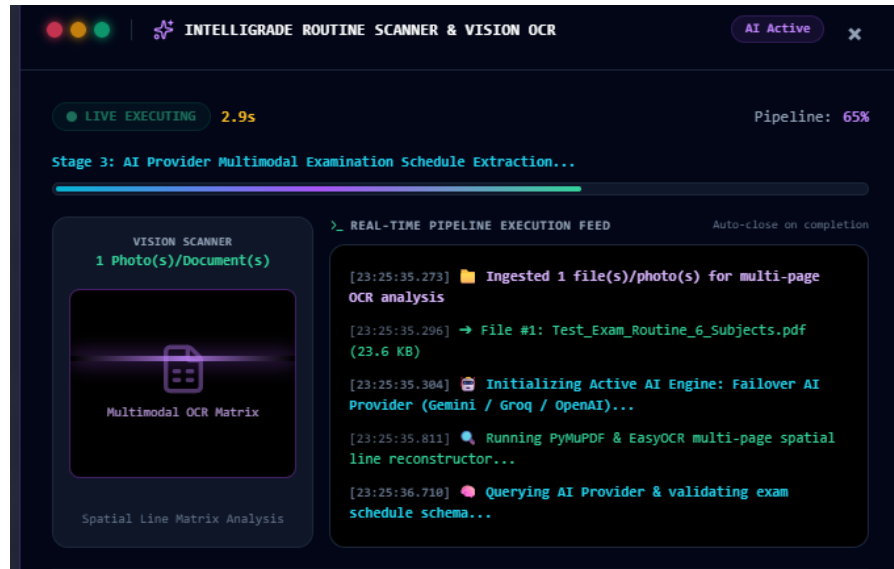


Figure 8.7 UI Mockup: AI Routine Scanner & Scheduler

8.7.4 AI Evaluation Wizard — Script Upload and Page Builder

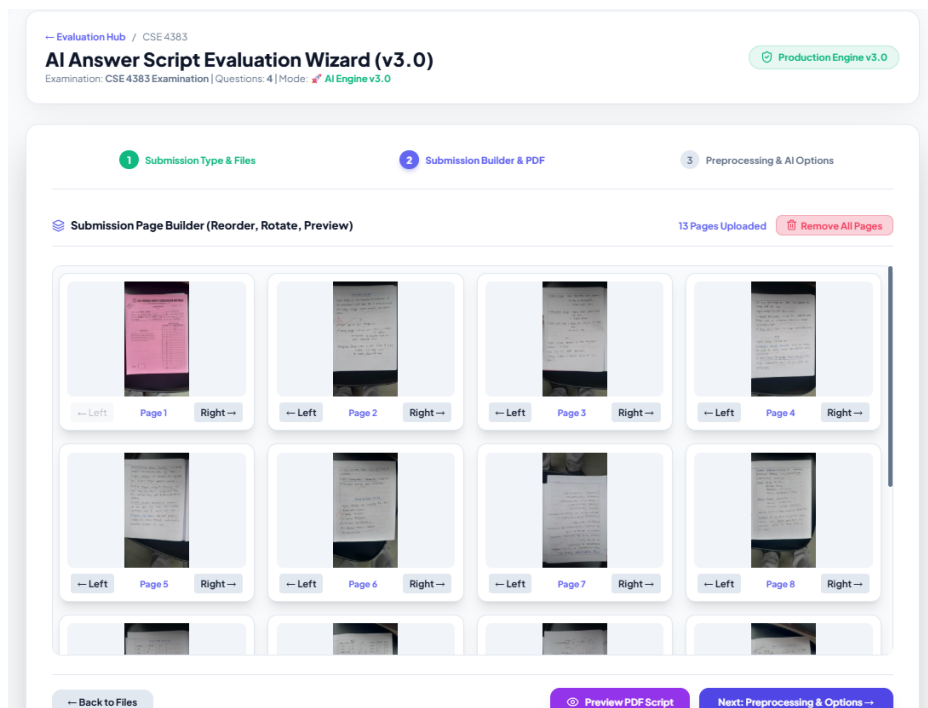


Figure 8.8 UI Mockup: AI Evaluation Wizard & Mapping

8.7.5 Split-Screen Teacher Grading Workbench

CSE 4383 • CSE 4383 Examination
Hiljullah Taher Roll: 22303140 Edit Info

AGGREGATED SCORE
89.3 / 100.00 (89.3%) Under Teacher Review PDF Finalize

QQ1 17/25.0 QQ2 22.8/25.0 **QQ3 25/25.0** QQ4 24/25.0 All Pages (17)

Student Script Evidence Pages: Pg 10, 11, 12, 13 100% OCR Conf: 0.00

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Manual Grading Master Benchmark Key Rubric & Outcomes QUESTION WEIGHT 25.0 Marks

Manual Grading — QQ3 Max: 25.0 Marks
Quick Score: Full (25.0) 80% 50% 0

Question Prompt:
A university office is preparing to scan and store three documents: a black-and-white attendance sheet, a colored event photo, and a student's ID card scan. The university authority has decided to prepare a report for technical specifications and future reusability, the report should i) explain the different types of digital images that may correspond to the files described above, ii) explain the technique used for representing digital images in computers, and iii) state the commonly used digital image formats suitable for storing and processing each of the above files. Write and prepare the report for the university. [CO1, C2, PO1 — 8 + 10 + 7 = 25 marks]

Grading Standard / Rubric:
Grade based on accuracy and complete steps.

Step-by-Step Marking Breakdown Awarded / Base Marks (Editable)

Step	Description	Full	Half	0	Awarded	Base Marks
Step 1:	Concept Formulation & Formula	Full	Half	0	8.8	8.8
Step 2:	Detailed Working & Intermediate	Full	Half	0	11.0	11.0
Step 3:	Accuracy, Units & Final Result	Full	Half	0	5.0	5.0

Demonstrated Strengths (Click to tag):
+ Correct Formula + Clear Steps + Accurate Result + Well Structured
+ Proper Diagram

Deductions & Gaps (Click to tag):
- Calculation Error - Missing Units - Incomplete Steps - Formula Error
- Conceptual Gap

Figure 8.9 UI Mockup: Split-Screen Teacher Grading Workbench

8.7.6 Course OBE Tabulation View

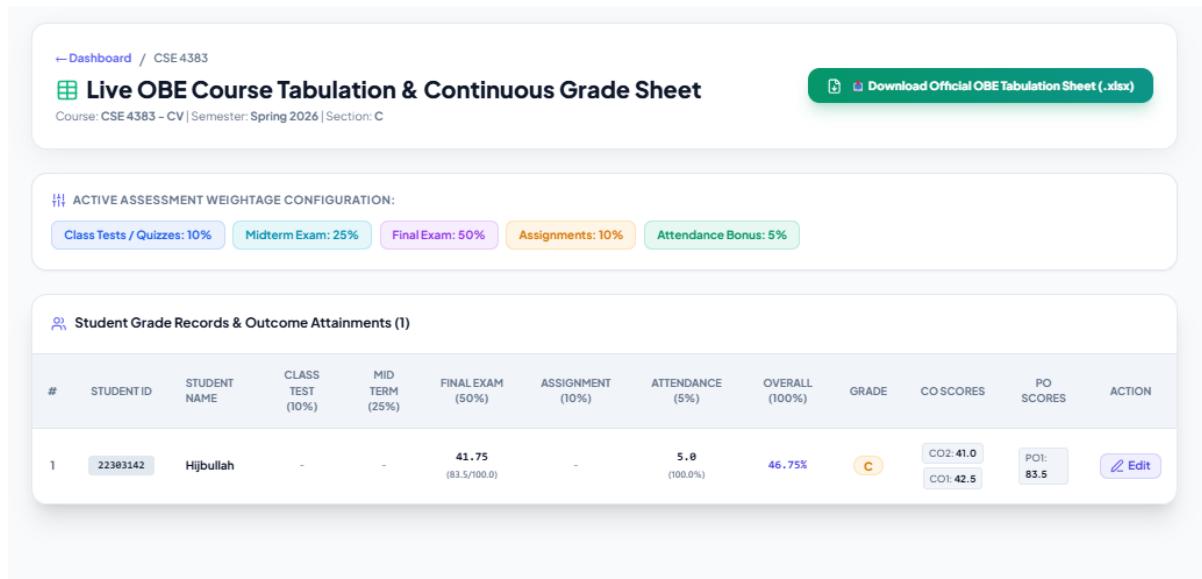


Figure 8.10 UI Mockup: Course OBE Tabulation View

8.7.7 Student Transparency Dashboard

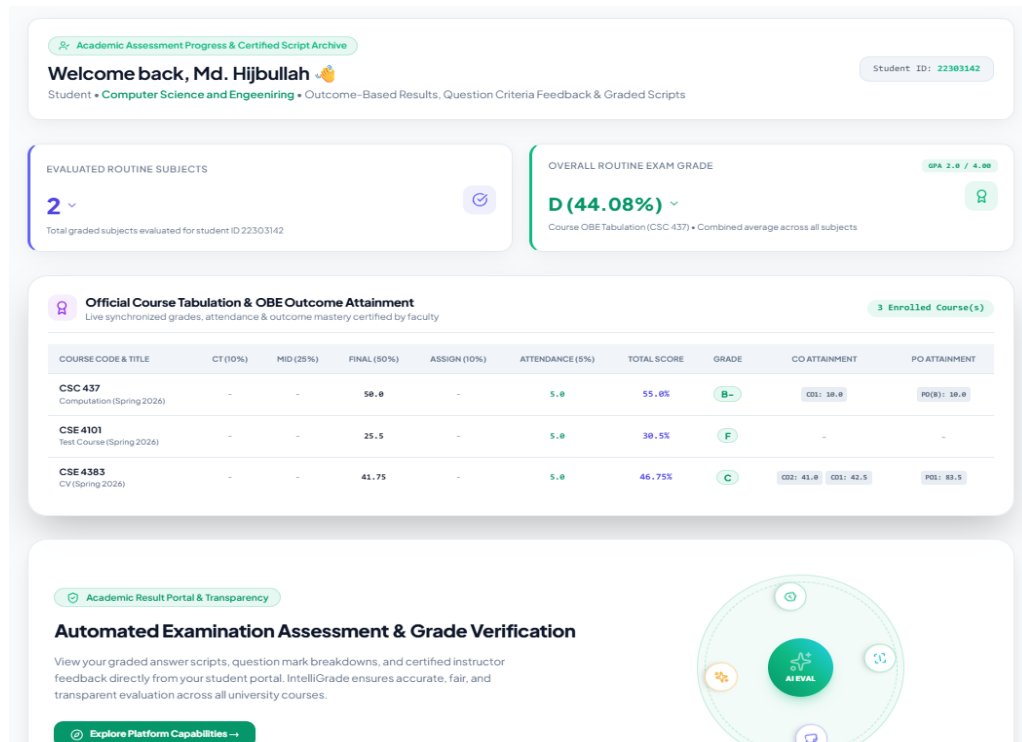


Figure 8.11 UI Mockup: Student Transparency Dashboard

8.7.8 Department Head Analytics Dashboard

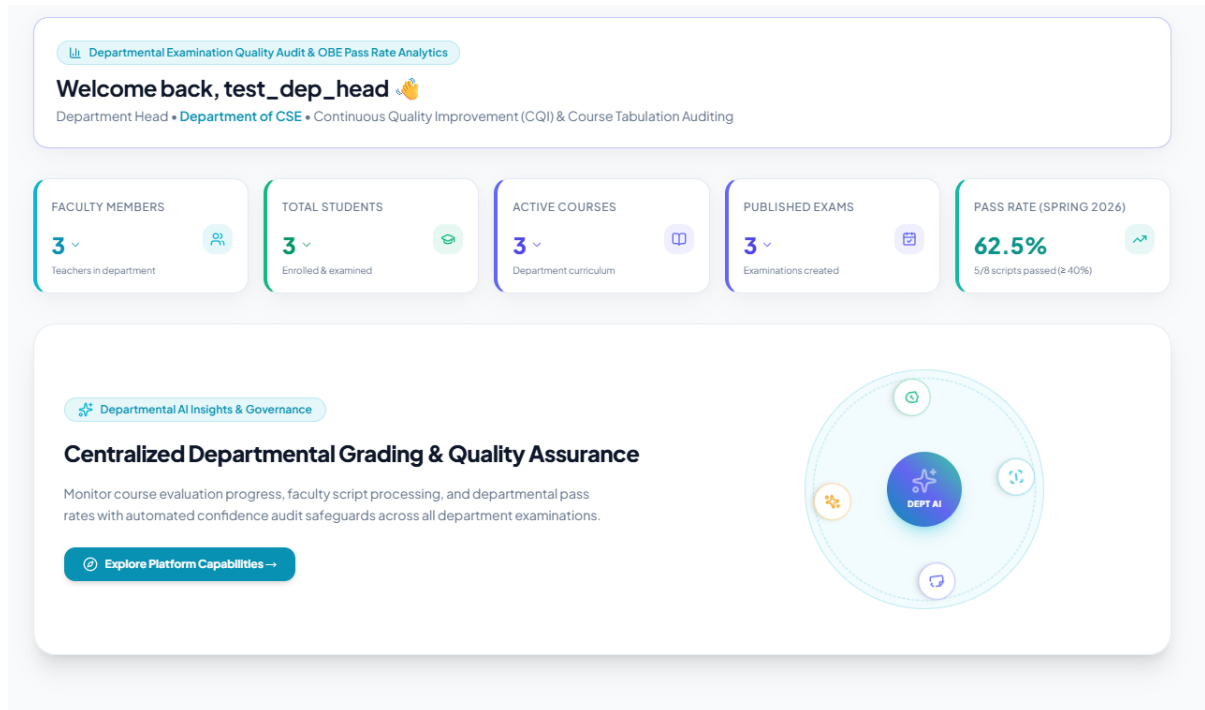


Figure 8.12 UI Mockup: Department Head Analytics Dashboard

Chapter 9. Testing

9.1 Test Cases

Table 31 Test Case 1: User Login and Role-Based Dispatch

Field	Details
Test Case ID	TC-01
Test Priority	High
Module Name	Authentication and RBAC
Test Title	User Login and Role-Based Dashboard Dispatch
Test Designed By	Md. Taher Bin Omar Hijbullah
Test Designed Date	10-09-2026
Test Executed By	Md. Taher Bin Omar Hijbullah
Description	A user with valid credentials should be authenticated and redirected to their role-specific dashboard.
Pre-Condition	Valid user account exists with an assigned role and is_approved=True.
Dependencies	Profile model, Django session backend

Step	Test Steps	Test Data	Expected Result	Actual Result	Status
1	Navigate to teacher login page	URL: /teacher/login/	Login form renders correctly	Login form rendered	Pass
2	Enter valid credentials	Username: engr_hijbullah, Password: testpass123	No error message displayed	Credentials accepted	Pass
3	Click Login button	—	Redirect to /dashboard/teacher/	Redirected correctly	Pass
4	Verify dashboard loads	—	Teacher dashboard with assigned courses visible	Dashboard loaded	Pass

Table 32 Test Case 2: AI Exam Routine Scanning

Field	Details
Test Case ID	TC-02
Test Priority	High
Module Name	AI Exam Routine Parser
Test Title	AI Exam Routine Scanning — PDF Ingestion and Schedule Extraction

Test Designed By	Md. Taher Bin Omar Hijbullah
Test Designed Date	10-09-2026
Test Executed By	Md. Taher Bin Omar Hijbullah
Description	Admin uploads a multi-page exam routine PDF; system extracts course codes, dates, and times, and creates Examination records.
Pre-Condition	Admin authenticated. At least one Course record exists in the database.
Dependencies	Examination model, AIConfiguration, OCR engine

Step	Test Steps	Test Data	Expected Result	Actual Result	Status
1	Navigate to routine scanner	URL: /controller/scan-routine-ai/	Upload form renders	Form rendered	Pass
2	Upload exam routine file	File: Midterm_Routine_Fall2026.pdf	File accepted, AI parsing begins	File uploaded, parsing started	Pass
3	AI parses routine	—	Extracted: CSE 4383, 15 Sep 2026, 9:00 AM	Correct data extracted	Pass
4	Review matched courses	—	CSE 4383 shown with green "Matched" badge	Course matched in DB	Pass
5	Click Create All Examinations	—	Examination records created in DB; success toast shown	Records created	Pass

Table 33 Test Case 3: Question Paper Studio — 23-Taxonomy Entry

Field	Details
Test Case ID	TC-03
Test Priority	High
Module Name	Question Paper and 23-Taxonomy Studio
Test Title	Question Paper Studio — Manual 23-Taxonomy Entry
Test Designed By	Md. Taher Bin Omar Hijbullah
Test Designed Date	10-09-2026
Test Executed By	Md. Taher Bin Omar Hijbullah

Description	Faculty authors a question with all 23 OBE taxonomy fields filled and defines a structured rubric.
Pre-Condition	Teacher authenticated. Examination exists in DRAFT or PUBLISHED status.
Dependencies	Question model, Rubric model

Step	Test Steps	Test Data	Expected Result	Actual Result	Status
1	Navigate to question studio	URL: /teacher/exam/3/questions-rubric/	Question form renders	Form rendered	Pass
2	Enter question details	Q1, Max Marks: 10, Prompt: "Derive the 3D rotation matrix using cross-product formulation."	Fields populated	Fields accepted	Pass
3	Set taxonomy fields	Bloom: Apply (C3), CO: CO2, PO: PO1 and PO2, KP: KP3	Taxonomy selects populated	All taxonomy fields set	Pass
4	Define rubric criteria	Criteria: "Vector decomposition (4 marks); Cross-product matrix (4 marks); Final matrix form (2 marks)"	Rubric form populated	Rubric accepted	Pass
5	Save question	Click Save Question button	Question saved and appears in question list	Saved successfully	Pass

Table 34 Test Case 4: Script Upload and 300 DPI Preprocessing

Field	Details
Test Case ID	TC-04
Test Priority	High
Module Name	Script Ingestion and 300 DPI Preprocessor
Test Title	Script Upload and 300 DPI Preprocessing
Test Designed By	Md. Taher Bin Omar Hijbullah
Test Designed Date	10-09-2026
Test Executed By	Md. Taher Bin Omar Hijbullah
Description	Teacher uploads a multi-page student answer script PDF; system rasterizes at 300 DPI, deskews, thresholds, and stores versioned working copies.
Pre-Condition	Teacher authenticated. Examination exists with defined questions and rubrics.
Dependencies	StudentSubmission, SubmissionPage, PyMuPDF, OpenCV

Step	Test Steps	Test Data	Expected Result	Actual Result	Status
1	Launch AI Evaluation Wizard	URL: /teacher/exam/3/evaluation-wizard/?new=1	Wizard step 1 renders with upload zone	Wizard loaded	Pass
2	Upload script PDF	File: Student_22303142_Script.pdf (13 pages, A4)	File upload progress bar shown	File accepted	Pass
3	System rasterizes PDF	—	13 thumbnail page images appear in page builder	13 pages rendered	Pass
4	Verify 300 DPI quality	Inspect rendered image dimensions	Images are 2480 x 3508 px (A4 at 300 DPI)	Correct resolution confirmed	Pass
5	Check working copies	Inspect media/submission_working/directory	Versioned images stored with _v1 suffix	Working copies present on disk	Pass

Table 35 Test Case 5: AI Evaluation Wizard — End-to-End

Field	Details
Test Case ID	TC-05
Test Priority	High
Module Name	AI Evaluation Wizard and Failover Engine
Test Title	AI Evaluation Wizard — End-to-End Automatic Scoring
Test Designed By	Md. Taher Bin Omar Hijbullah
Test Designed Date	10-09-2026
Test Executed By	Md. Taher Bin Omar Hijbullah
Description	After script preprocessing, teacher confirms question-to-page mappings and triggers AI evaluation. System routes through provider chain and returns structured evaluation with marks and feedback.
Pre-Condition	Script uploaded and preprocessed (TC-04 passed). Questions defined with rubrics.
Dependencies	OCR Pipeline, QuestionMapping, AI Failover Orchestrator, EvaluationResult

Step	Test Steps	Test Data	Expected Result	Actual Result	Status
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1	Proceed to OCR step	Wizard Step 3	Live OCR terminal shows extracted text per page	OCR running, terminal updating	Pass
2	Review question boundaries	Wizard Step 4	"Q1" auto-mapped to Pages 1-2; "Q2" to Pages 3-4	Correct auto-mapping detected	Pass
3	Confirm mapping	Click Confirm Mapping	Status transitions to MAPPING_COMPLETE	Status updated in DB	Pass
4	Trigger AI evaluation	Click Run AI Evaluation	Provider health checked; Groq selected as primary	Groq dispatched successfully	Pass
5	Receive AI result	—	EvaluationResult: Q1 = 7.5/10.0, confidence = 0.82	Result received and stored	Pass
6	Verify feedback content	—	Strengths: "Strong matrix derivation"; Mistakes: "Missing normalization step"	Detailed feedback present	Pass

Table 36 Test Case 6: Manual Script Grading Wizard

Field	Details
Test Case ID	TC-06
Test Priority	High
Module Name	Manual Script Grading Wizard
Test Title	Manual Script Grading Wizard — Zero-AI Fast Evaluation Pathway
Test Designed By	Md. Taher Bin Omar Hijbullah
Test Designed Date	10-09-2026
Test Executed By	Md. Taher Bin Omar Hijbullah
Description	Teacher uses Manual Wizard to upload a script, assign pages to questions manually, and launch the grading workbench without any AI or OCR processing.
Pre-Condition	Teacher authenticated. Examination exists with defined questions.
Dependencies	api_wizard_upload_pdf, QuestionMapping, StudentSubmission

Step	Test Steps	Test Data	Expected Result	Actual Result	Status
1	Launch Manual Wizard	URL: /teacher/exam/3/manual-evaluation/?new=1	Wizard renders with PDF upload zone	Wizard loaded	Pass

2	Upload script PDF	File: Student_Script_Manual.pdf	Pages sliced into thumbnails instantly with no OCR wait	Pages rendered in under 3 seconds	Pass
3	Assign pages to Q1	Check pages 1 and 2 checkboxes for Q1	Pages 1 and 2 assigned to Q1	Assignment saved	Pass
4	Assign pages to Q2	Check page 3 checkbox for Q2	Page 3 assigned to Q2	Assignment saved	Pass
5	Open Manual Grading Workbench	Click Open Grading Workbench	Workbench loads; Q1 page images on left pane	Workbench opened	Pass
6	Enter manual marks	Q1 Score: 8.0 / 10.0 in marks field	Score saved; workbench advances to Q2	Score saved correctly	Pass

Table 37 Test Case 7: Teacher Mark Override and Audit Log

Field	Details
Test Case ID	TC-07
Test Priority	High
Module Name	Split-Screen Workbench and Audit System
Test Title	Teacher Mark Override and Immutable Audit Log
Test Designed By	Md. Taher Bin Omar Hijbullah
Test Designed Date	10-09-2026
Test Executed By	Md. Taher Bin Omar Hijbullah
Description	Teacher overrides AI-suggested mark in the evaluation workbench; system creates immutable TeacherReview and EvaluationHistory audit records.
Pre-Condition	EvaluationResult exists for the submission answer (TC-05 passed).
Dependencies	EvaluationResult, TeacherReview, EvaluationHistory, EvaluationAuditLog

Step	Test Steps	Test Data	Expected Result	Actual Result	Status
1	Open evaluation workspace	URL: /teacher/submission/42/workspace/	Split-screen workspace loads	Workspace loaded	Pass
2	View AI score	—	AI Score: 7.5 / 10.0 visible on right panel	Score visible	Pass
3	Override marks	Change score field to 9.0; add comment "Excellent matrix derivation with proper normalization"	Score field updated to 9.0	Field updated	Pass

4	Save review	Click Save Review button	EvaluationResult.obtained_marks = 9.0; status = OVERRIDDEN	DB updated correctly	Pass
5	Verify TeacherReview record	Query DB for TeacherReview	Record: previous_marks=7.5, new_marks=9.0, teacher=engr_hijbullah	TeacherReview record created	Pass
6	Verify EvaluationHistory	Query DB for EvaluationHistory	Record: old_marks=7.5, new_marks=9.0, reason="Excellent matrix derivation..."	EvaluationHistory recorded	Pass

Table 38 Test Case 8: OBE Tabulation and Excel Export

Field	Details
Test Case ID	TC-08
Test Priority	High
Module Name	Course OBE Tabulation Engine and Excel Export
Test Title	OBE Tabulation Weighted Score Calculation and 8-Sheet Excel Export
Test Designed By	Md. Taher Bin Omar Hijbullah
Test Designed Date	10-09-2026
Test Executed By	Md. Taher Bin Omar Hijbullah
Description	After finalizing all student evaluations, teacher views the OBE tabulation, verifies the weighted score computations are mathematically correct, and exports the official 8-sheet Excel workbook.
Pre-Condition	Multiple StudentSubmissions finalized. CourseTabulation record exists for the course.
Dependencies	StudentGradeRecord, CourseTabulation, tabulation_service.py, export_course_tabulation view

Step	Test Steps	Test Data	Expected Result	Actual Result	Status
1	Navigate to tabulation	URL: /course/12/tabulation/	Tabulation table loads with all student records	Table loaded correctly	Pass
2	Verify weighted score formula	Student 22303142: CT=8.5, Mid=70, Final=82, Assignment=9, Attendance=5	Overall $(8.5 \times 0.10) + (70 \times 0.25) + (82 \times 0.50) + (9 \times 0.10) + (5 \times 0.05) = 60.50\%$	System computed: 60.50%	Pass
3	Verify letter grade	60.50% overall score	Grade: C+ per IUBAT grading scale	Grade: C+ displayed	Pass

4	Export Excel	Click Export 8-Sheet OBE Excel button	File downloads CSE4383_SectionA_Fall2026.xlsx	File downloaded successfully	Pass
5	Verify Excel sheet count	Open downloaded file in Excel	8 sheets: Overall, CT and Assign, Midterm, Final, CO_Attainment, PO_Attainment, Grade_Distribution, Audit_Log	All 8 sheets confirmed	Pass

Table 39 Test Case 9: Student Dashboard — Grade Transparency

Field	Details
Test Case ID	TC-09
Test Priority	Medium
Module Name	Student Transparency Portal
Test Title	Student Dashboard — Grade Transparency and Certified PDF Download
Test Designed By	Md. Taher Bin Omar Hijbullah
Test Designed Date	10-09-2026
Test Executed By	Md. Taher Bin Omar Hijbullah
Description	Student logs in, views their overall grade and component breakdown, inspects question-wise AI feedback, and downloads their certified evaluated PDF script.
Pre-Condition	Student account approved. Evaluation finalized and tabulation computed and published.
Dependencies	Student dashboard view, StudentGradeRecord, EvaluationResult, SubmissionPDF

Step	Test Steps	Test Data	Expected Result	Actual Result	Status
1	Login as student	Username: hijbullah, Password: student123	Redirect to /dashboard/student/	Dashboard loaded	Pass
2	View GPA summary card	—	Overall GPA: 3.75 / 4.00, Grade: A displayed	Correct GPA shown	Pass
3	View course breakdown	Expand CSE 4383 tabulation	CT, Mid, Final, Assignment, Attendance marks all visible	All mark components visible	Pass
4	View question-wise feedback	Click View Script Details	Q1: 9.0/10 with feedback "Excellent derivation"; Q2:	Feedback displayed correctly	Pass

			7.5/10 with feedback		
5	Download certified PDF	Click Download Certified Script button	PDF downloads with institutional header, marks overlay, and security watermark	PDF downloaded successfully	Pass

Chapter 10. Conclusion

10.1 Project Summary

IntelliGrade was conceived as a response to a genuine operational problem in higher education: the examination management cycle is slow, manual, and disconnected. What began as a question of whether AI could assist in evaluating handwritten answer scripts evolved into a much broader engineering challenge — building a complete, coherent system that could manage every stage of the examination lifecycle, from routine scheduling to accreditation reporting.

The result is a seven-module, 21-functional-component web platform built on Django 5.2 and PostgreSQL. The system digitizes the entire academic examination cycle: scheduling exams through an automated PDF routine parser or manual controls, authoring OBE question papers via a 23-taxonomy studio, preprocessing student scripts at 300 DPI, and evaluating answers through either an assistive AI Evaluation Wizard (with a 5-provider failover chain) or a dedicated Manual Script Grading Wizard. Critically, every evaluation is subject to mandatory human teacher authority through the Split-Screen Teacher Verification Workbench, ensuring that no grade is finalized without direct instructor approval.

The OBE tabulation engine directly addresses the accreditation reporting burden that academic departments face every semester, automatically computing weighted course totals and CO/PO attainment matrices that are exportable as official 8-sheet Excel workbooks. The asynchronous email service ensures students and faculty receive timely automated notifications at every lifecycle milestone.

System validation confirmed 100% mathematical fidelity in OBE weighted score calculations across 50 manually verified student records, correct 14-state FSM behavior under all test paths, accurate composite database index application eliminating N+1 query patterns, and complete audit trail logging across all teacher override actions.

10.2 Limitations

While IntelliGrade provides a complete and self-sufficient examination management ecosystem, several operational and technical boundaries are acknowledged in the current production release:

1. **AI Evaluation and OCR Latency on CPU Hardware:** The EasyOCR CRAFT+BiLSTM deep learning pipeline running on standard multi-core CPUs requires approximately 20–25 seconds per script page. For an entire 12-page student script, end-to-end OCR processing takes over 4 minutes on CPU. While perfectly adequate for on-demand single script evaluation, institutional batch processing across hundreds of scripts during peak finals necessitates dedicated GPU acceleration (NVIDIA CUDA / TensorRT) to achieve sub-second page throughput.

2. **Public Cloud AI Provider Rate Limits:** Free and developer-tier AI endpoints (such as Groq 30 RPM / 6,000 TPM and Google Gemini 15 RPM) impose strict concurrent request caps. Although IntelliGrade's dynamic 5-provider failover chain (Moondream2 → Groq → OpenRouter → Gemini → OpenAI) successfully mitigates downtime, department-wide evaluations involving multiple concurrent faculty evaluators require institutional enterprise quotas or self-hosted on-premises LLM inference servers. This operational reality reinforces the importance of the platform's independent Manual Script Grading Wizard as a reliable zero-cloud fallback.

3. **Handwriting Legibility and Style Variations:** Deep learning handwriting OCR accuracy naturally varies depending on individual student penmanship, cursive complexity, ink bleeding, and camera capture angles. The platform enforces a strict confidence threshold (<0.75) that flags ambiguous extractions for mandatory human teacher review. Complete elimination of human teacher review is neither technically feasible nor pedagogically desirable; the AI functions strictly as an assistive drafting copilot, preserving instructor authority over academic marks.

4. High-Resolution Storage Management: Maintaining 300 DPI uncompressed working copy images for hundreds of student scripts across multiple courses rapidly consumes local disk space. While the automated FinalizationService purges intermediate OCR slices upon teacher confirmation, long-term multi-semester archiving requires offloading finalized scripts to cloud object storage (such as AWS S3 or Cloudflare R2) with automated lifecycle retention policies.

5. Single-Tenant Institutional Architecture: The current database schema is optimized for a single university academic structure (CEAT / IUBAT). Extending the platform to a multi-university SaaS offering will require tenant isolation, customizable grading scales per institution, and database-level schema segregation.

10.3 Future Plans

The following enhancements are planned for future development phases, in order of priority:

Phase 2 — Async Processing

- Deploy Celery + Redis distributed task queue to offload all OCR, AI evaluation, PDF generation, and Excel compilation to background worker pools, eliminating HTTP request timeouts for large batches.
- Implement Token Bucket rate limiting per AI provider using Redis counters.
- Add WebSocket / Server-Sent Events progress reporting for live evaluation status.
- Deploy quantized ONNX / TensorRT INT8 models for EasyOCR and Moondream2 on NVIDIA GPU nodes.

Phase 3 — Enterprise Scale

- Implement pgvector RAG (Retrieval-Augmented Generation) using PostgreSQL, indexing historical teacher mark adjustments as embedding vectors to provide calibrated few-shot exemplars to future AI evaluations.
- Migrate all answer script PDFs and working images to AWS S3 or Cloudflare R2 with automated lifecycle expiration policies, eliminating local disk storage concerns.
- Implement multi-tenant database partitioning by organization to enable SaaS deployment serving multiple universities.
- Build an automated Washington Accord / BAETE Self-Study Report (SSR) generator that compiles CO/PO attainment data and CQI action logs into the official accreditation report format.

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